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United States Patent Application Publication

20250276056

Kind Code

A1

Publication Date

September 04, 2025

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CYTOLYTIC T CELL IMMUNOTHERAPY FOR HIGHLY PATHOGENIC CORONAVIRUSES

Abstract

Compositions and methods to induce cytolytic T lymphocytes (CD8+) response, that is, MHC class I restricted T cell responses, to pathogenic and common cold coronaviruses, including a delivery platform for antigens consisting of a polyionic papillomavirus virus-like particle (VLP), with contiguous, negatively charged amino acids flanked by a cysteine residue inserted in the HI loop of the papillomavirus L1 protein. Antigens to be paired with the VLP include fusion peptide/proteins derived from a pathogenic coronavirus, and from the genetically most closely related human coronaviruses that commonly circulate in human populations, with N-terminal or C-terminal amino acids consisting of contiguous, positively charged amino acids preceded and/or followed by a cysteine residue and a C-terminal proteolytic processing sequence (AAYY) to enhance presentation of MHC class I epitopes.

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Family ID: 82447540

Appl. No.: 18/272331

Filed (or PCT Filed): January 14, 2022

PCT No.: PCT/US2022/012434

Related U.S. Application Data

us-provisional-application US 63137402 20210114

Publication Classification

Int. Cl.: A61K39/215 (20060101); A61K39/00 (20060101); A61P31/14 (20060101)

U.S. Cl.:

CPC A61K39/215 (20130101); A61P31/14 (20180101); A61K2039/5258 (20130101); A61K2039/572 (20130101); C12N2710/14023 (20130101); C12N2710/14034 (20130101); C12N2770/20034 (20130101)

Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present invention relates to compositions and methods for inducing cellular immune responses to coronaviruses.

Description of the Background

[0002] Coronaviruses (CoVs) are enveloped viruses with a single-strand, positive-sense RNA genome approximately 26-32 kilobases in size. SARS-CoV-2, like all coronaviruses, shares common features in the organization and expression of its genome; nonstructural proteins encoded by open reading frame (ORF) 1a/b, are followed by the principal structural proteins spike(S), envelope (E), membrane (M), and nucleocapsid (N) (Chen et al., 2020). The S glycoprotein mediates attachment to the host receptor. The S glycoprotein is cleaved by a host protease into two separate polypeptides designated S1 and S2. S1 makes up the receptor binding domain of the S protein and is the principal target of neutralizing antibodies, while S2 forms the stalk of the spike molecule. The M protein is the most abundant viral protein and participates in the formation of the core viral particle. The N protein is the only nucleocapsid protein and forms the portion of the core particle that interacts with the viral genome. The CoVs are separated into four genera based on phylogeny: α -CoV, β -CoV, γ -CoV and δ -CoV. Within the beta-CoV genus, four lineages (A, B, C, and D) are recognized.

[0003] Coronaviruses cause a large variety of diseases in animals. In humans, CoV infections primarily involve the upper respiratory tract and the gastrointestinal tract, and principally cause mild, self-limiting disease, such as the common cold. Two of these human coronaviruses are α -coronaviruses, HCoV-229E and HCoV-NL63, while the other two are β -coronaviruses, HCoV-OC43 and HCoV-HKU1. HCoV-229E and HCoV-OC43 were isolated nearly 50 years ago, while HCoV-NL63 and HCoV-HKU1 have only recently been identified. These viruses are endemic in the human populations, causing 15-30% of respiratory tract infections each year. Seroprevalence studies suggest that exposure to these viruses is nearly universal in humans (Severance et al., 2008). The existence of additional human coronaviruses is plausible but unknown. The first highly pathogenic human coronavirus was SARS-COV, a β -coronavirus. It was identified as the causative agent of the Severe Acute Respiratory Syndrome (SARS) outbreak that occurred in 2002-2003 in the Guangdong Province of China. A second novel, highly pathogenic human CoV emerged in the Middle East in 2012. This virus, named Middle East Respiratory Syndrome-CoV (MERS-CoV), is also a β -coronavirus and was found to be the causative agent of highly pathogenic respiratory tract infections in Saudi Arabia and other countries in the Middle East. The third and most recent highly pathogenic human coronavirus, SARS-CoV-2, first appeared in China in late 2019, is also a β -coronavirus, and is currently responsible for an ongoing pandemic. The highly pathogenic coronaviruses are believed to circulate in zoonotic reservoirs, principally in bat species, with

occasional spillover into the susceptible human population, possibly via an intermediate host species. The occurrence of 3 cross species transmission events during the past 17 years raises the prospect that similar viruses may emerge in the future. The defining feature of highly pathogenic coronaviruses is their ability to cause serious morbidity and mortality in infected individuals, with crude estimates of 2.3%, 9.5%, and 34% for SARS-CoV-2, SARS-CoV-1 and MERS, respectively (Petrosillo et al., 2020). The commonly circulating human coronaviruses rarely if ever cause serious morbidity or mortality.

[0004] Treatment of pathogenic coronaviruses is primarily supportive. Despite ongoing efforts, there are no highly active anti-viral drugs.

SUMMARY OF THE INVENTION

[0005] The present invention relates to treatment and prevention of disease caused by the highly pathogenic coronaviruses, SARS-CoV-1, MERS and SARS-CoV-2, and other related pathogenic coronaviruses.

[0006] The approach of the present invention to treatment of coronaviruses is an immunotherapy which harnesses the natural immune defenses of the body. For clearance of an existing viral infection the paradigm of immune defense is the MHC class 1-restricted CD8⁺ cytotoxic T lymphocyte, which has the ability to destroy and clear virally infected cells. However, it is problematic whether induction of T cells directed against an infecting virus can clear that infection before the host is overwhelmed, because a maximal response to immunization typically requires weeks, perhaps months to achieve. Therefore, in the case of highly pathogenic coronaviruses, a strategy is needed to rapidly mobilize an efficacious anti-viral cytolytic T cell response to control or clear acute infection. The proposed approach is to stimulate cross reactive memory T cells. Memory T cells, which are more responsive to stimulation than naïve cells, can be clonally expanded very rapidly. Recent studies have identified cross-reactive T cells that recognize SARS-CoV-2 antigens in blood samples obtained prior to the appearance of the virus in 2019 (Grifoni et al., 2020; Weiskopf et al., 2020; Braun et al., 2020; Le et al., 2020). Since the common cold coronaviruses, OC43 and HKU1, share amino acid similarities with SARS-CoV-2, these cross-reactive T cells may be induced by prior infections with common cold human coronaviruses. In fact, a recent study that mapped the epitopes recognized by cross reactive T cells demonstrated that pre-existing memory CD4⁺ T cells are cross reactive with SARS-CoV-2 and common cold coronaviruses (Mateus et al., 2020). The detailed molecular and structural basis for cross reactive T cells is poorly understood but the current paradigm of immune recognition predicts the requirement for highly conserved amino acid sequences within MHC class I-restricted T cell epitopes (contiguous sequences 8-13 amino acids in length) that are shared by SARS-CoV-2 and the common cold coronaviruses. Alignments of SARS-CoV-2 amino acid sequences with homologous regions of known common human coronaviruses show only limited regions of high amino acid identity in putative T cell epitopes. However, the notion that cross reactivity requires substantial amino acid identity between epitopes from heterologous viruses is based on the clonal selection theory of immune recognition, which postulates individual lymphocytes are specific for a single antigen, one clone-one specificity, and requires close amino acid identity between cross reactive epitopes. Several scientists have called this theory into question and proposed a theory of T cell immune recognition that postulates T cells recognize multiple specificities, one clone-millions of specificities (Mason, 1998; Wilson et al., 2004; Kersh and Allen, 1996; Sewell, 2012). The theory is grounded in the mathematic consideration that there are only 10^{12} T cells in humans and $<10^{15}$ distinct T cell receptors in the human naïve T cell pool, while the theoretical limit of possible peptides of 20 amino acids that can bind to MHC molecules is vast ($>10^{15}$) and likely even greater if peptides that contain post-translational modifications are considered. This theory of T cell cross reactivity leads to a novel approach to both treat and prevent highly pathogenic coronaviruses. Simultaneous induction of T cell responses to a highly pathogenic coronavirus and to one or more genetically related common human coronaviruses to which the majority of

individuals have prior exposure and thus have memory T cells will generate two categories of cellular immunity, therapeutic and prophylactic. The strategy will recall cross-reactive partially or fully protective memory T cells and induce naïve T cells specific for the highly pathogenic coronavirus. The dual action of these two classes of T cell response will be therapeutic by rapidly controlling and eradicating the acute highly pathogenic coronavirus infection in the host, preventing severe symptoms and death, and at the same time be prophylactic by providing long term protection against future infection with the highly pathogenic coronavirus.

[0007] The present invention further relates to compositions and methods to induce cytolytic T lymphocytes (CD8+) response, that is, MHC class I restricted T cell responses, to pathogenic and common cold coronaviruses.

[0008] The invention relates to a delivery platform for antigens consisting of a polyionic papillomavirus virus-like particle (VLP), wherein contiguous, negatively charged amino acids flanked by a cysteine residue are inserted in the HI loop of the papillomavirus L1 protein.

[0009] The invention further relates to antigens to be paired with the VLP comprising fusion peptide/proteins with N-terminal or C-terminal amino acids consisting of contiguous, positively charged amino acids preceded and/or followed by a cysteine residue, here after designated the TAG. The invention further relates to the TAG having a C-terminal proteolytic processing sequence (AAYY) to enhance presentation of MHC class I epitopes.

[0010] In particular embodiments, the invention relates to antigens that are derived from a pathogenic coronavirus, and from the genetically most closely related human coronaviruses that commonly circulate in human populations. The invention further relates to the choice of antigens for induction of a cytolytic T cell response based on the abundance of expression of the viral protein in virally infected cells, the density and position of predicted MHC class I-restricted T cell epitopes in the protein, and empirical studies identifying cytolytic T cells directed toward specific viral proteins. The invention further relates to the choice of antigens comprising two separate and distinct classes of targets that are derived from viral structural proteins and viral non-structural proteins, respectively. Since the first proteins expressed in virally infected cells are non-structural viral proteins, targeting these proteins provides for an efficacious cellular immune response at the earliest stage of viral infection. The invention further relates to the choice of antigens of the common nonpathogenic human coronavirus based on the above criteria and the additional criterion that the antigen encompasses regions of the viral protein that share at least 40% identity to the corresponding viral protein of the pathogenic coronavirus. In particular embodiments of VLP-antigen compositions, the antigens are either short peptides 25-30 amino acids in length, extended peptides approximately 45-55 amino acids in length, short proteins approximately 110-140 amino acids in length, or full-length amino acid sequences of target antigens.

[0011] In a particular embodiment, the pathogenic coronavirus is SARS-CoV-2 and the genetically related human coronaviruses are OC43 and HKU1. In a particular embodiment the antigens of SARS-CoV-2 are derived from the following viral structural proteins, the membrane protein (M), the nucleocapsid protein (N), ORF3a, ORF7a, and the S2 region of the spike(S) envelope protein and the antigens of the non-structural proteins are derived from the nsp6, nsp7 and nsp12. In a particular embodiment, the antigens of the structural proteins of OC43 and HKU1 are derived from the M and N proteins and the S2 region of the S protein, and the antigens of the non-structural proteins are derived from the nsp3, nsp4, nsp6, nsp7 and nsp12 proteins of OC43, and variable regions of HKU1 with respect to OC43 from the same viral proteins.

[0012] The invention also relates to the method of preparing polyionic papillomavirus VLPs paired with coronavirus antigens. The invention further relates to the method of delivery of the VLP-antigen composition by the intravenous, intramuscular or intradermal route. In a particular embodiment, the VLP-antigen composition is delivered intranasally or by inhalation to stimulate lung and nasopharyngeal tissue resident memory T cells and generate cytotoxic T lymphocytes that preferentially traffic to sites into the respiratory tract.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] FIG. 1 shows alignment of bovine papillomavirus (BPV) type 1 L1 protein with the L1 protein of the closely related BPV isolate NY8385: HI loop is aa344-357.

[0014] FIG. 2 shows alignment of M proteins of OC43 and SARS-CoV-2.

[0015] FIG. 3 shows alignment of M proteins of OC43 and HKU1.

[0016] FIGS. 4A and 4B show alignments of antigenic regions of N proteins of OC43 and SARS-CoV-2.

[0017] FIGS. 5A, 5B and 5C show alignment of antigenic regions of N proteins of OC43 and HKU1.

[0018] FIGS. 6A and 6B show alignments of antigenic regions of S proteins of OC43 and SARS-CoV-2 within the S2 region.

[0019] FIG. 7 shows alignment of S proteins of HKU1 and OC43 within a variable antigenic region.

[0020] FIGS. 8A, 8B and 8C show alignment of antigenic regions of ORF1ab of OC43 with SARS-CoV-2 nsp3 protein.

[0021] FIGS. 9A, 9B and 9C show Alignment of antigenic regions of ORF1ab of OC43 with SARS-CoV-2 nsp4 protein.

[0022] FIGS. 10A and 10B show alignment of antigenic regions of ORF1ab of OC43 with SARS-CoV-2 nsp6 protein.

[0023] FIG. 11 shows alignment of antigenic region of ORF1ab of OC43 with SARS-CoV-2 nsp7 protein.

[0024] FIG. 12 shows the sequence of Bovine Papillomavirus type 1 capsid protein L1.

[0025] FIG. 13A shows alignment of a first set of antigenic regions of ORF1ab of OC43 with SARS-CoV-2 nsp12 protein.

[0026] FIG. 13B shows alignment of a second set of antigenic regions of ORF1ab of OC43 with SARS-CoV-2 nsp12 protein.

[0027] FIG. 13C shows alignment of a third set of antigenic regions of ORF1ab of OC43 with SARS-CoV-2 nsp12 protein.

DETAILED DESCRIPTION

[0028] The present invention provides compositions and methods relating to genetically engineered papillomavirus L1 virus-like particles (VLPs) comprising negatively charged amino acid sequences and a cysteine residue in the HI loop of the L1 protein, hereafter referred to as polyionic VLPs. The invention further relates to an L1 protein from any papillomavirus species, including human, bovine, equine, murine, ovine, porcine, cervine, canine, feline, or leporine. In one embodiment, the L1 papillomavirus protein is from bovine papillomavirus type 1 (BPV1) (FIG. 12).

[0029] In particular embodiments, the HI loop of the genetically engineered BPV1 L1 protein comprises 4 to 10 contiguous, negatively charged amino acids, flanked by a cysteine residue at the N- or C-terminus or both termini. The HI loop of BPV1 L1 is here defined as amino acid positions 344 to 357, with a span of 14 amino acids and the tip defined as the proline (P) residue at position 349 (FIG. 1). The negatively charged amino acids of the VLPs of the present invention can be glutamic acid, aspartic acid, or both. In various embodiments, the amino acid sequence of 5 to 12 amino acids in length (4-10 contiguous, negatively charged amino acids and one or two flanking cysteine residues) may be inserted in the HI loop, and replace none, or one, or more than one of the native amino acids. In specific exemplary embodiments, the amino acid sequence inserted in the HI loop is 9 residues in length, comprising 8 negatively charged amino acids, glutamic acid or aspartic acid or alternating aspartic and glutamic acids, and a C- or N-terminal cysteine residue, and replaces the 9 native amino acids at positions 347-355 of the HI loop of the bovine papillomavirus

L1 protein (Table 1, inserts 1-4). In other specific exemplary embodiments, the amino acid sequence inserted in the HI loop is 5, 6, 7, 8, 10, or 11 amino acids in length, comprising 4, 5, 6, 7, 9, or 10 glutamic acids, respectively, and a C-terminal cysteine residue. The respective inserts replace the equivalent number of native amino acids at positions, 347-351, 347-352, 347-353, 347-354, 346-355 or 345-355, respectively, of the HI loop of the bovine papillomavirus L1 protein (Table 1 insert 5-10). In a specific exemplary embodiment, the amino acid sequence inserted in the HI loop is 9 residues in length, comprises 8 negatively charged glutamic acids and a C- and N-terminal cysteine residue, and replaces the 10 native amino acids at positions 346-355 of the HI loop of the bovine papillomavirus L1 protein (Table 1, insert 11). In other various embodiments the negatively charged amino acids and cysteine(s) may be inserted in the HI loop and replace fewer native amino acids than the number comprising the insert. In specific exemplary embodiments a 9-amino acid sequence, comprising 8 glutamic acids and a C-terminal cysteine residue replaces 7, 5 or 3 native amino acids at positions 348-354, or 348-352, or 348-350, respectively, of the HI loop of the bovine papillomavirus L1 protein (Table 1, inserts 12-14). In other various embodiments the negatively charged amino acids and cysteine(s) may be inserted in the HI loop and replace up to 2 more native amino acids than the number comprising the insert. In a specific exemplary embodiment, a 9-amino acid sequence, comprising 8 glutamic acids and a C-terminal cysteine residue, replaces the 11 native amino acids at position 346-356 of the HI loop of the bovine papillomavirus L1 protein (Table 1, insert 15). In various embodiments, the glutamic acid-cysteine amino acid sequence inserted in the HI loop may replace the proline at the putative tip of the loop or may be inserted between the proline residue and the immediate C-terminal leucine residue without removing any native amino acids. In some embodiments, the inserted amino acid replacing proline or located between the proline and leucine residues may be flanked by a glycine-serine-serine-glycine (GSSG) linker amino acid sequence. In specific exemplary embodiments, the amino acid sequence inserted in the HI loop comprises 8 glutamic acids and a C-terminal cysteine residue, with or without flanking GSSG amino acids, and either replaces the proline at position 349 or is located between amino acid positions 349 and 350 amino acid of the HI loop of the bovine papillomavirus L1 protein (Table 1, inserts 16-19).

[0030] The invention further relates to antigens to be linked to the VLP, comprising fusion peptide/proteins with N-terminal or C-terminal amino acids consisting of 4-10 contiguous, positively charged amino acids preceded by and/or followed by a cysteine residue, here after designated the TAG. The TAG allows the antigen to be attached to the polyionic papillomavirus VLP by the combined action of electrostatic interactions and an oxidization reduction reaction between cysteine residues on the VLP and the cysteine residue in the TAG. The invention further relates to the TAG having a C-terminal proteolytic processing sequence (AAYY) to enhance presentation of MHC class I epitopes. Where the TAG is appended to the C-terminus of the antigen, the proteolytic processing sequence (AAYY) is placed at the N-terminus of the peptide/protein antigen.

[0031] In specific exemplary embodiments the TAG comprises the group of positively charged amino acids, arginine (R), lysine (K) or histidine (H), and the sequence is 8 amino acids in length, (Table 2, Tag-1, -2, -3). In other specific exemplary embodiments, the TAG comprises a repeating motif of RKHRKHRK, 8 amino acids in length (Table 2, TAG-4). In other specific exemplary embodiments, the TAG comprises an amino acid sequence of 4, 5, 6, or 7 contiguous arginines followed by a cysteine residue (Table 2, TAG-5, -6, -7, and -8) or an amino acid sequence of 9 or 10 arginines followed by a cysteine residue (Table 2 TAG-9 and -10). In other specific exemplary embodiments, the TAG consists of 8 positively arginines preceded by a cysteine residue or flanked at the N and C terminus by cysteine residues (Table 2, TAG-11 and -12)

[0032] The invention further relates to the composition of polyionic papillomavirus VLPs and target antigens with a TAG sequence for the purpose of inducing cytolytic (MHC class I restricted) T cell responses. In specific embodiments, the invention relates to coronavirus antigens linked to

polyionic VLPs for the purpose of inducing cytolytic T cell responses to common cold coronaviruses and pathogenic coronaviruses.

[0033] The entire proteome of a virus can be a target of T cell responses. However, structural proteins are particularly effective targets for antigen-specific T cell responses because they are the most abundantly expressed viral proteins in infected cells, thus allowing for efficient presentation to MHC molecules. Empirical studies of cellular immune responses to coronaviruses also support the importance of structural proteins as principal targets of the cellular immune response (reviewed in (Liu et al., 2017)). Cytolytic T cell responses are directed toward MHC class I restricted T cell epitopes. These epitopes are most commonly 9 amino acids in length and can range from 8-13 amino acids in length. The genetic heterogeneity of MHC class I alleles in humans and other mammals allows for numerous possible epitopes within a viral protein. In order to encompass the universe of possible epitopes, an antigen needs to be the full-length amino acid sequence of a target protein or sets of shorter fragments of the antigen that together include the entire amino acid sequence. The preferred length of antigenic fragments linked to polyionic VLPS is an empirical function of efficiency of epitope presentation for induction of a T cell response via the alternative antigen presentation pathway. In addition, manufacturing considerations may influence the preferred choice of length of an antigen.

[0034] The preferred embodiment for a polyionic papillomavirus VLP composition of matter for the induction of cytotoxic T cell response to SARS-CoV-2 comprises antigens from the membrane (M), nucleocapsid (N), S2 region of the spike(S), ORF3a, and ORF7a structural proteins and the nsp6, nsp7 and nsp12 nonstructural proteins.

[0035] The M protein is 222 amino acids in length and the antigenic region spans aa6-221. The N protein is 419 amino acids in length. Based on empirical studies and predictive algorithms for MHC class I-restricted T cell epitope the antigenic region is aa51-369. The Spike(S) protein is 1270 amino acids in length and comprises two regions, S1 (aa1-661) containing the receptor binding domain (aa330-583) and S2 (aa662-1270). T cell epitopes are widely distributed across the S protein. The S1 region contains the majority of virion surface exposed amino acids and is the principal target of humoral immune responses, including neutralizing antibodies. The region is preferably excluded from a vaccine intended to induce cellular immunity to avoid unintended induction of antibody responses with possible deleterious effects, such as induction of antibodies mediating antibody dependent enhancement (ADE). ORF3a is 275 amino acids in length. Empirical studies support the choice of ORF3a as a target antigen for T cell response. Prediction algorithms (MHC-NP: prediction of peptides naturally processed by MHC, developed by Sébastien Giguère, Alexandre Drouin, Alexandre Lacoste, Mario Marchand, Jacques Corbeil and François Laviolette; <http://tools.iedb.org/mhcnp/>) show that the C terminus, as compared to the N terminus, has a greater density of putative MHC class I restricted epitopes for 5 representative alleles (67 spanning the C-terminal 171 amino acids versus 20 spanning the 104 N-terminal amino acids). Empirical studies support choice of ORF7a, a protein of 121 amino acids, as a target of cytolytic T cell responses. The ORF1ab of SARS-CoV-2 is a polyprotein of 7096 amino acids. The polyprotein is proteolytically processed within virally infected cells to yield multiple non-structural proteins that serve diverse functions in the viral life cycle. The proteins are present in low abundance in virally infected cells and for this reason are not major targets of cellular immune responses, but recent studies have shown that cytolytic T cell responses to several nsp proteins can be detected in the blood of SARS-CoV-2 infected patients and also in blood obtained from some healthy blood donors in the pre-COVID19 era (Grifoni et al., 2020; Le et al., 2020). Because these proteins are the first to be expressed in infected cells they are attractive targets for cytolytic T cells. Rapid killing of these virally infected cells can prevent the cell from producing infectious virus. In terms of frequency of antigen specific T cells and percentage of responding subjects, the principal targets of CD8+ T cell responses are the nsp6, nsp7 and nsp12 proteins.

[0036] The amino acid length of antigens linked to polyionic VLPs can range from a defined

epitope of 8-14 amino acids or longer amino acid sequences up to the full-length amino acid sequence of the antigenic region of a viral protein or fusions of several viral proteins. In exemplary embodiments, the antigens are short peptides approximately 25-30 amino acids in length, extended peptides approximately 45-55 amino acids in length, short proteins approximately 110-140 amino acids in length, or the full-length amino acid sequence of the antigenic region of a targeted viral protein. For antigenic fragments shorter than the full-length amino acid sequence of the antigenic region, the specific embodiment comprises a set of antigens that includes all the amino acids of the antigenic region of the target viral protein. Antigens for the M protein of SARS-CoV-2, embodied as short peptides, extended peptides, short proteins, or the entire amino acid sequence of the antigenic region, are shown in Table 3. Comparable antigens for the N protein of SARS-CoV-2 are shown in Table 4. Comparably designed antigens for the S2, ORF3a, and ORF7a structural proteins of SARS-CoV-2 are shown in Tables 5-7. Antigens for the nonstructural proteins nsp6 and nsp7 are shown in Table 8 and antigens for the nsp12 protein are shown in Table 9 and 10. Short and extended peptides are overlapping by 10 amino acids and short protein antigens are overlapping by 11 amino acids. In the final formulation, peptide, protein, or full-length antigens have a TAG, as described above.

[0037] The preferred embodiment for a polyionic papillomavirus VLP composition of matter to stimulate memory T cells induced by prior exposure to common cold coronaviruses comprises antigens derived from both structural and nonstructural proteins. The multiple specificities theory of T cell recognition does not define at the molecular level the structural basis for T cell cross reactivity. We define herein viral proteins or sub regions of viral proteins containing cross-reactive T cell epitopes as those with an average amino acid identity of >40% across homologous amino acid sequences of a common cold coronavirus and a specific pathogenic coronavirus.

[0038] In particular embodiments, the common coronaviruses are OC43 and HKU1, and the antigens for induction of cross-reactive T cell responses are from the M, N and S2 structural proteins and the nsp3, nsp4, nsp6, nsp7 and nsp12 nonstructural proteins. Where the amino acids sequences of OC43 and HKU1 share on average >40-80% identity across homologous amino acids of the target antigen, only the amino acid sequences of OC43 are used as the target antigens. In other particular embodiments, the amino acid sequences of HKU1 are the antigens.

[0039] The M protein of OC43 is 230 aa in length and shares overall identity of 40.8% with the M protein of SARS-CoV-2 within the region of aa14-226 (FIG. 2). Alignment of OC43 M protein with that of HKU1 shows no significant areas of variability (amino acid identity <80%) (FIG. 3). The N protein of OC43 is 448 aa in length and on average shares 36% amino acid identity with the N protein of SARS-CoV-2. The antigenic region of OC43 from aa99-400, excluding non-aligned regions aa266-269, aa341-349, and aa382-387, shares 43.5% identity with the homologous region of the N protein of SARS-CoV-2, after excluding aa221-224 that do not align with the amino acid sequence of the OC43 N protein (FIG. 4A). The N-terminus amino acids of the OC43 N protein at positions aa64-88 share 52% identity with the corresponding region of N of SARS-CoV-2 (FIG. 4B). Alignment of OC43 N protein with that of HKU1 shows 3 antigenic regions with low levels of amino acid identity between the two viral amino acid sequences, ranging from 33%-50%. (FIGS. 5A, 5B, and 5C). The S protein of OC43 is 1353 amino acids and composed of 2 sub-regions, S1 and S2. The S1 extends from aa 1-789 and shows low amino acid identity (24%) with the S1 region of SARS-CoV-2 in the aligned region between aa70-552. In contrast, the S2 region of the viruses contains two antigenic regions, aa898-1153 and aa1228-1302, that share an average of 52.72% and 51.3% amino acid identity, respectively (FIG. 6). Alignment of the homologous region of HKU1 with aa898-1153 of OC43 showed overall identity of 83.5% without regions of high variability, while the antigenic region of OC43 between aa1228-1303 shares 70.7% identity with the homologous S2 region of HKU1 (FIG. 7).

[0040] In exemplary embodiments, the antigens are short peptides approximately 25-30 amino acids in length, extended peptides approximately 45-55 amino acids in length, short proteins

approximately 110-140 amino acids in length, or the full-length amino acid sequence of the antigenic region of a targeted viral protein. For antigenic fragments shorter than the full-length amino acid sequence of the antigenic region, the specific embodiment comprises a set of antigens that includes all the amino acids of the antigenic region of the target viral protein. Antigens for the M protein of OC43 and variable regions of HKU1, embodied as short peptides, extended peptides, short proteins, or the entire amino acid sequence of the antigenic region, are shown in Table 11. Comparable antigens for the N protein of OC43 and variable regions of HKU1 are shown in Table 12, and antigens for the S2 protein of OC43 and variable regions of HKU1 in Table 13. Short and extended peptides are overlapping by 10 amino acids and short protein antigens are overlapping by 11 amino acids. In the final formulation peptide, protein or full-length amino acid sequences include a TAG, as described above.

[0041] The ORF1ab of OC43 coronaviruses encodes a polyprotein of 7095 amino acids. The polyprotein is proteolytically processed within virally infected cells to yield multiple non-structural proteins that serve diverse functions in the viral life cycle. In terms of frequency of antigen specific T cells and percentage of responding subjects, the principal targets of CD8⁺ T cell responses are the nsp3, nsp4, nsp6, nsp7 and nsp12 proteins. The choice of antigens from these nonstructural proteins is also based on the additional consideration of an average of >40% identity between amino acid sequences of common cold viruses and SARS-CoV-2.

[0042] The nsp3 protein is 1,945 amino acids in length. The overall identity between SARS-CoV-2 and OC43 nsp3 amino acids sequences is 26%. However, the C-terminal region of 384 amino acids shares 39% identity and contains within it 3 regions of continuous amino acids with 52.8%, 41%, and 43.6% identity between the amino acid sequence of OC43 and the homologous amino acid sequence of SARS-CoV-2 (FIG. 8A-C). The nsp 4 protein is 500 amino acids in length; the nsp4 proteins of SARS-CoV-2 and OC43 share 42% amino acid identity. Three sub-regions in the mid portion or C terminus of the nsp4 protein, which are 45, 133 and 50 amino acids in length, share 53.3%, 47.4% and 72% amino acid identity, respectively, between homologous amino acid sequences of SARS-CoV-2 and OC43 (FIG. 9A-C). The nsp6 protein is 290 amino acids in length and the homologous amino acid sequences of SARS-CoV-2 and OC43 share 30.5% identity. The C-terminal 84 amino acids share 54.8% identity and a 31 amino acid region at the N-terminus shares 41.9% identity (FIG. 10A-B). The nsp7 protein is 83 amino acids in length and the amino acid sequences of the SARS-CoV-2 and OC43 nsp7 share 46% identity. Excluding the C-terminal 13 amino acids of SARS-CoV-2, the amino acid identity of SARS-CoV-2 and OC43 nsp7 proteins is 55.2% (FIG. 11). The nsp12 protein is 932 amino acids in length. An empirical study of antigen specific cellular immune responses in SARS-CoV-2 infected patients located the majority of T cell epitopes in the regions encompassed by aa 125-375 and aa 520-920 of the protein (Grifoni et al, 2021) Two amino acid sequences within OC43 ORF1ab share 61.5% and 67% identity with corresponding amino acids within the aa125-275 fragment of the SARS-CoV-2 nsp12 protein (FIGS. 13A-B) and 341 aa of OC43 ORF1ab shares 76.8% identity with corresponding amino acids within the aa520-920 fragment of the SARS-CoV-2 nsp12 protein (FIG. 13C). The nonstructural proteins of OC43 and HKU1 share ~90% identity and have no extended regions of amino acid variability (identity<80%).

[0043] In exemplary embodiments, the antigens are short peptides approximately 25-30 amino acids in length, extended peptides approximately 45-55 amino acids in length, short proteins approximately 110-140 amino acids in length, or the full-length amino acid sequence of the antigenic region of a targeted viral protein. For antigenic fragments shorter than the full-length amino acid sequence of the antigenic region, the specific embodiment comprises a set of antigens that includes all the amino acids of the antigenic region of the target viral protein. Antigens for the nsp3 and nsp4 proteins, embodied as short peptides, extended peptides, short proteins, or the entire amino acid sequence of the antigenic region are shown in Table 14. Comparably designed antigens for the nsp6 and nsp7 proteins of OC43 are shown in Table 15 and for nsp12 in Tables 16 and 17.

Short and extended peptides are overlapping by 10 amino acids and short protein antigens are overlapping by 11 amino acids. In the final formulation peptide, protein or full-length amino acid sequences include a TAG, as described above.

EXAMPLES

Example 1: Generation of Recombinant Baculoviruses with Genetically Engineered BPV L1 Genes
[0044] The entire open reading frame (ORF) of BPV L1 with a Kozak consensus and unique restrictions sites at each end (EcoR1/Not1) is artificially engineered by PCR-based gene synthesis and cloned in a pUC18 vector. The entire ORF is codon-modified using *Drosophila melanogaster* preferred codons, for efficient expression in insect cells. The ORF contains insertion of peptides with aspartic or glutamic acid residues and a cysteine residue and inserted into the HI loop as described in Table 1, and designated insert-1 to insert-19.

[0045] The modified BPV L1 genes are subcloned between the EcoR1/Not1 sites of the pORB baculovirus transfer vector. The transfer vectors are co-transfected with a linear baculovirus DNA in *Spodoptera frugiperda* sf9 cells using a preferred commercially available transfection reagent, as suggested by the manufacturer. Five days post-transfection, the recovered recombinant baculoviruses are further amplified by large scale infections of sf9 cells. Small scale infections to confirm expression of the modified L1 proteins are conducted with 2×10^6 *Trichoplusia ni* (High Five) cells, growing in 6-well plates and infected with serial dilutions of Baculovirus stocks. 72 hrs post-infection, the cells are lysed in 500 μ l of RIPA buffer and the clarified lysates are subjected to SDS-PAGE analysis to detect overexpression of a protein of the expected molecular weight of 55 kDa.

Example 2: Production of Polyionic VLPs from Recombinant Baculoviruses

[0046] For production of VLPs, approximately 2×10^9 *Trichoplusia ni* (High Five) cells growing in spinner flasks are infected with a pre-determined amount of a high-titer recombinant baculovirus stock in 500 ml of TNM-FH/10% FBS. After 96 h of incubation at 27° C., the cells are harvested, and collected by centrifugation at 2,000 rpm (Sorvall FH18/250 rotor) for 5 min. Cell pellets are resuspended in extraction buffer (20 mM phosphate buffer, pH 6.5, 1 M NaCl, 0.1 mM CaCl₂, 50 μ M FeCl₂) containing protease inhibitors (Roche Complete ULTRA, 1 tablet per 10 ml) and subjected to 5 cycles of thawing at 37° C. and freezing in a -80° C. ethanol bath. The lysate is spun 1 h at 8000 rpm to remove baculovirus particles. The clarified lysate is extracted for 10 min with an equal volume of Vertrel DF (Fisher Scientific). The aqueous layer is loaded onto a 40% sucrose cushion and centrifuged in a SW32Ti rotor at 32,000 rpm for 1.5 h. The sucrose pellet is resuspended in 20 mM phosphate pH 8.0, 0.5 M NaCl, 5 mM MgCl₂, and incubated 30 min at 37° C. with 250 U/ml of Salt Active Nuclease (Arcticzymes). After dialysis in 20 mM phosphate pH 6.5, 0.5 M NaCl, the VLP solution is adjusted to 0.01% Tween 80, 0.05% carboxymethyl cellulose, 50 μ M FeCl₂ and stored at 4° C. Purity is assessed by SDS-PAGE gel analysis and protein concentration is measured by Bradford dye method and uv spectroscopy. To facilitate direct visualization of VLPs, an aliquot of diluted particles is placed on 300-mesh formvar/carbon-coated copper grids, negatively stained with 2% phosphotungstic acid (pH=7.0) and examined by transmission electron microscopy (TEM). Chimeric VLPs with inserts from Table 1 are shown to yield capsid-like structures ~50 nm in diameter with a variegated appearance consistent with formation from smaller capsomere-like structures and resembling native HPV type 1 VLPs.

Example 3: Conjugation of Antigens to Polyionic VLPs

[0047] For conjugation to the VLP, peptides are solubilized in distilled water at 5 mg/ml (or dissolved at 5 mg/ml in DMSO, if they are not soluble in water). Peptides at a preferred concentration of 2.5 mg/ml, but lower if less soluble in the buffer, are reduced with 10 mM Bond-breaker TCEP solution (Thermo Fisher Scientific) for 20 min at 50° C. After dialysis in 20 mM phosphate buffer, pH 6.5, 0.15 M NaCl, VLP protein (1 mg/ml), and peptide at peptide: L1 protein molar ratios between 4:1 to 16:1, based on solubility characteristics, are mixed in the presence of 4 mM glutathione disulfide (GSSG) and 0.8 mM reduced glutathione (GSH), and incubated

overnight at 37° C. To remove unreacted peptide, reactants are dialyzed against 20 mM phosphate pH 6.5, 0.5 M NaCl, using dialysis tubing with a cut-off of 1 million kDa. The VLP-peptide solutions are adjusted to 0.01% Tween 80, 0.05% carboxymethyl cellulose, 0.5 mM GSSG and 0.05 mM GSH, aliquoted, and stored at -20° C. The amount of peptide bound to the VLP is determined by SDS-PAGE analysis and interpolation of sample-peptide band density from a standard curve of known amounts of peptide. Gels are scanned in a BioRad ChemiDocXR imager, and images are analyzed with NIH ImageJ software.

Example 4: Ability of Exemplary Embodiments of VLPs with Various Inserts in the HI Loop to Induce a Cytolytic CD8+ T Cell Response

[0048] VLPs generated with various exemplary inserts in the L1 protein (Table 1) are linked via a TAG to a representative peptide encoding an MHC class I restricted epitope recognized in the genetic background of C57BL/6 mice. Mice are immunized by the intradermal route, as described below. The immune response is measured as described below. VLPs formulated with different inserts linked to a representative antigen are shown to induce comparable frequencies of antigen specific, interferon- γ secreting CD8+ T cells (no significant differences in mean responses by t-test comparison) cytolytic T cell responses.

Example 5: Ability of VLPs Linked to an Antigen by Exemplary Embodiments of TAGs to Induce a Cytolytic CD8+ T Cell Response

[0049] A representative peptide encoding an MHC class I-restricted epitope recognized in the genetic background of C57BL/6 mice is chemically synthesized to >90% purity with the TAGs listed in Table 2. Peptides with TAGs-1 to -4, and -11 and -12 are linked to a polyionic VLP with Insert-1. Peptides with TAGs 5-10 are linked to VLPs with inserts 5-10, respectively. Mice are immunized by the intradermal route, as described below. The immune response is measured as described below. VLPs linked to a representative antigen formulated with different TAGs are shown to induce comparable frequencies of antigen specific, interferon- γ secreting CD8+ T cells (no significant differences in mean responses by t-test comparison).

Example 6: Immunization with Polyionic VLPs and Detection of Antigen Specific CD8+ T Cell Responses

[0050] C57BL/6J mice 6-8 weeks of age are immunized three times, 1 week apart with between 5-50 ug each of VLP-peptide, administered by intradermal injection. As a control, C57BL/6J mice are immunized with unlabeled (no peptide) VLP protein. VLP-peptide immunogens are formulated with the set of extended peptides for the difference SARS-CoV-2 and OC43 antigens described in Table 3-12. For intradermal immunization, the VLP/antigen vaccine is injected in a single, or split doses, into the skin of the back of shaved mice.

[0051] To provide samples for immunological assays, mice are sacrificed 10-14 days after the last dose of vaccine and spleens are dissected. Splenocytes (or suspensions of CD3+ cells from lung tissue), are stimulated with 1 μ g/ml of a pool of overlapping peptides 11 amino acids in length (overlapping by 8 amino acids) spanning the full-length amino acid sequence of the target antigen, in the presence of brefeldin A (10 ug/ml) overnight at 37° C. in 5% CO₂. Cells are stained with Zombie Green™ fixable viability dye, treated with Fixation buffer and stored in Cytolast. Cells are permeabilized with Permeabilization buffer and stained with Brilliant Violet BV™ 510-conjugated anti-mouse CD3, clone 17A2, PerCPCy5.5-conjugated anti-mouse CD8 α , clone 53-6.7, and PE-conjugated anti-mouse IFN γ , clone XMG1.2. Reagents are purchased from a commercial source. Flow cytometry is performed on an LSR-II or comparable flow cytometer and data are analyzed using FACSDiva software or FlowJo software. Gating is done on forward and side scatter parameters to select for lymphocytes and singlets. After exclusion of dead cells, CD8+T lymphocytes are identified on a CD3/CD8 dot plot of gated lymphocytes, and interferon- γ (IFN γ) secreting cells are identified on a CD8/IFN γ dot plot of gated CD8+ T cells. A minimum of 30,000 CD8+ T cells are analyzed. VLP-peptide immunogens of the M, N, S2, ORF3a, ORF7a, nsp6, nsp7, and nsp12 antigens of SARS-CoV-2, and immunogens of the M, N, S2, nsp6, nsp7, and

nsp12 antigens of OC43 are shown to induce detectable antigen specific, interferon- γ secreting CD8+ T cells

Example 7: Ability of Polyionic VLP SARS-CoV-2 Vaccines to Protect Mice Against SARS-CoV-2 Disease in a Challenge Model

[0052] Mice strain: Stable humanized angiotensin converting enzyme-II (ACE2) mice generated using CRISPR/Cas9 knock-in technology to replace the endogenous mouse ACE2 (mACE2) with the human ACE2 in the C57BL/6 strain of mouse will be used for SARS-CoV-2 challenge (Sun et al., 2020).

[0053] Mouse immunization: VLPs with an exemplary insert in the HI loop are formulated with the set of exemplary extended peptides with an exemplary and appropriate TAG for the particular insert. A polyionic VLP vaccine is formulated with M, N, S2, ORF3a, ORF7a, nsp6, nsp7 and/or nsp12 SARS-CoV-2 antigens of SARS-CoV-2 antigens. Peptides are drawn from the list in Tables 3-10 with SEQ ID ending in extension, -E1, -E2, -E3, etc. The precise number of peptides will depend on the antigen as described in the table. The preferred insert is the E8C amino acid sequence at amino acid positions 347 to 355 in the HI loop of the L1 protein of bovine papillomavirus type 1, with replacement of native amino acids (insert-1, Table 1), and the preferred TAG is CRRRRRRRRCAAYY (TAG-1, in Table 2). Additional insert and TAG combinations with sets of peptides for one or more SARS-CoV-2 antigens are also tested, as informed by other enabling experiments. VLP/antigen constructs are generated as described above. Mice are immunized by the intradermal route, intranasal/lung route, or both routes simultaneously. For intradermal immunization, the VLP/antigen vaccine is injected in a single, or split doses, into the skin of the back of shaved mice. For nasal/pulmonary immunization, the VLP/antigen construct is administered dropwise (10 μ l) into the nose of a lightly anesthetized mouse. In anesthetized mice intra-nasally administered vaccine is also inhaled by the mouse and thus is delivered to both the lung and nasopharyngeal tissues.

[0054] Mouse challenge: For intranasal infection, aged (30 weeks old) hACE2 mice are anesthetized with Isoflurane delivered with a precision vaporizer, and then intranasally infected with 4×10^5 pfu of SARSCoV-2. Mice are then weighed and monitored daily and sacrificed on day 6 post infection for serum collection and tissue processing. Spleens and lung tissue are collected for analysis of viral RNA load, histopathology, and measurement of cellular immune response.

[0055] Measurement of viral RNA load: Viral RNA in lung tissue is extracted with a RNeasy Mini kit (QIAGEN) according to the protocols. The viral RNA quantification is performed by RT-qPCR targeting the S gene of SARS-CoV-2. RT-qPCR is performed using One Step PrimeScript RT-PCR Kit (Takara) with the following primers and probes: CoV-F3, CoV-R3 and CoV-P3 (Sun et al., 2020).

[0056] Antigen specific CD8+ T cell responses: CD8+ T cell response of splenocytes and CD3+ T cells recovered from lung homogenates are measured as described above.

[0057] SARS-CoV-2 polyionic VLP vaccines are shown to induce antigen specific CD8+ T cells responses detectable in splenocytes and in CD3+ T cells from lung tissue, to each of the target antigen (M, N S2, ORF3a and ORF7a). Vaccinated mice are further shown to have a significantly reduced viral RNA load and less lung pathology than mock vaccinated mice. Vaccines are also shown to decrease expression of viral RNA in the lung. Of note, the vaccines are not expected to provide sterilizing immunity.

Example 8: Ability of Polyionic VLP SARS-CoV-2 Vaccines to Protect Syrian Hamsters Against SARS-CoV-2 Infection in a Challenge Model

[0058] Animal species: The Syrian hamster is highly susceptible to SARS-CoV-2 infection, making it the most suitable small animal model to evaluate the protective efficacy of vaccines (Rosenke et al., 2020; Chan et al. 2020). Syrian hamsters (*Mesocricetus auratus*), 6-8 weeks of age are purchased from Jackson Laboratories.

[0059] Hamster immunization: VLPs with an exemplary insert in the HI loop are formulated with

the set of exemplary extended peptides with an exemplary and appropriate TAG for the particular insert. A polyionic VLP vaccine is formulated with M, N, S2, ORF3a, ORF7a, nsp6, nsp7 and/or nsp12 SARS-CoV-2 antigens. Peptides are drawn from the list in Tables 3-10 with SEQ ID ending in extension, -E1, -E2, -E3, etc. The precise number of peptides will depend on the antigen as described in the table and may include all or fewer antigens with a-E extension designation. The preferred insert is the E8C amino acid sequence at amino acid positions 347 to 355 in the HI loop of the L1 protein of bovine papillomavirus type 1, with replacement of native amino acids (insert-1, Table 1), and the preferred TAG is CRRRRRRRRRCAAYY (TAG-1, in Table 2). Additional insert and TAG combinations with sets of peptides for one or more SARS-CoV-2 antigens are also tested, as informed by other enabling experiments. VLP/antigen constructs are generated as described above. Hamsters are immunized by the intradermal route, intranasal route, or both routes individually or in combination. Intradermal immunization is performed as described above. For nasal immunization, the VLP/antigen construct is administered dropwise (10 µl) into the nose of a lightly anesthetized mouse.

[0060] Hamster challenge: To mimic the natural route of infection, vaccinated animals and controls (contacts) will be exposed to previously infected animals (index) by co-housing in the same cage. For intranasal infection of the index animal, hamsters are anesthetized with Isoflurane delivered with a precision vaporizer, and then intranasally infected with 100 tissue culture dose 50 (TCID₅₀) of SARSCoV-2 virus. SARS-CoV-2 isolate nCoV-WA1-2020 (MN985325.1) from the CDC, or a suitable alternative isolate, is obtained and propagated in Vero E6 cells. The TCID₅₀ dose is determined by titration of the viral stock in VeroE6 cells. The infected hamster and co-housed vaccinated and control naïve hamsters are weighed and monitored daily for clinical signs of disease. To monitor infection by RT-qPCR, nasal washes are collected from lightly anesthetized naïve contact (vaccinated and control) and index (previously infected) animals daily for 10 days by instillation and then collection of 150 µl of PBS/0.3% BSA in both nostrils.

[0061] Measurement of viral RNA load: Viral RNA in lung tissue is extracted with a RNeasy Mini kit (QIAGEN) according to the protocols. The viral RNA quantification is performed by RT-qPCR targeting the S gene of SARS-CoV-2. RT-qPCR is performed using One Step PrimeScript RT-PCR Kit (Takara) with the following primers and probes: CoV-F3, CoV-R3 and CoV-P3 (Sun et al., 2020).

[0062] SARS-CoV-2 polyionic VLP vaccinated hamsters are shown to have a significantly reduced viral RNA load in nasal washes than mock vaccinated mice after exposure to an infected index hamster.

Tables

TABLE-US-00001

TABLE	1	Representative negatively charged amino acid- cysteine sequences in HI loop	Replaced Position(s) of Inserted aa	native aa replacement in Seq ID	sequence	BPV1	L1	ORF	Insert-1	EEEEEEEEEC	GTPLTEYDS	Aa347-355			
Insert-2	CEEEEEEE	GTPLTEYDS	Aa347-355	Insert-3	DDDDDDDDDC	GTPLTEYDS	Aa347-355	Insert-4	EDEDEDEDC	GTPLTEYDS	Aa347-355	Insert-5	EEEEEC	GTPLT	Aa347-351
Insert-6	EEEEEC	GTPLTE	Aa347-352	Insert-7	EEEEEEEC	GTPLTEY	aa347-353	Insert-8	EEEEEEEC	GTPLTEYD	Aa347-355	Insert-9	EEEEEEEEEEEC	DGTPLTEYDS	Aa346-355
Insert-10	EEEEEEEEEEEC	DGTPLTEYDSS	Aa346-356	Insert-11	CEEEEEEEEC	DGTPLTEYDS	aa346-355	Insert-12	EEEEEEEEEC	TPLTEYD	aa348-354	Insert-13	EEEEEEEEEC	TPLTE	aa348-352
Insert-14	EEEEEEEEEC	TPL	aa347-350	Insert-15	EEEEEEEEEC	DGTPLTEYDS	aa346-3556	Insert-16	EEEEEEEEEC	P	aa349	Insert-17	GSSGEEEEEEEE	P	aa349
Insert-18	EEEEEEEEEC	None	Between aa349-350	Insert-19	GSSGEEEEEEEE	none	Between	ECGSSG	aa349-350						

TABLE-US-00002

TABLE	2	Representative TAG sequences	Amino acid	Seq ID	TAG designation	sequence
sup.1	TAG-1	polyR8	RRRRRRRRRC	TAG-2	PolyK8	KKKKKKKKKC
	TAG-3	PolyH8	HHHHHHHHHC	TAG-4	MixedRKH8	RKHKRHRKC
	TAG-5	PolyR4	RRRRRC	TAG-6		

PolyR5 RRRR PolyR6 RRRR TAG-7 RRRR TAG-8 PolyR7 RRRR TAG-9 PolyR9
 RRRRRRRRRRC TAG-10 PolyR10 RRRRRRRRRRC TAG-11 NH-terminal C CRRRRRRRR
 TAG-12 Dual C CRRRRRRRRRC .sup.1All TAG amino acid sequences include a C-terminal
 AAYY proteolytic processing sequence.
 TABLE-US-00003 TABLE 3 SARS-CoV-2 antigens of the M structural protein.
 Immunogen Seq ID aa-position.sup.1 Amino acid sequence.sup.2 Short SARS-CoV-2-M-S1
 aa6-36 GTITVEELKKLLEQWNLVIGFLFTWIC peptide LLQ SARS-CoV-2-M-S2 aa27-56
 LFLTWICLLQFAYANRNRFLYIIKLIFLY WL SARS-CoV-2-M-S3 aa48-77
 IIKLIFLWLLWPVTLACFVLA AVYRIN WIT SARS-CoV-2-M-S4 aa68-97
 AVYRINWITGGIAIAMACLVGLMWL SYFI SARS-CoV-2-M-S5 aa88-117
 VGLMWLSYFIASFRLFARTRSMWSFNP ETN SARS-CoV-2-M-S6 aa108-138
 SMWSFNPETNILLNVPLHGTILTRP LLE SEL SARS-CoV-2-M-S7 aa129-159
 LTRP LLESELVIGAVILRGHLRIAGHHL GRC SARS-CoV-2-M-S8 aa150-179
 RIAGHHLGRCDIKDLPKEITVATSRTL S YY SARS-CoV-2-M-S9 aa170-200
 VATSRTL SYYKLGASQRVAGDSGFAA YSRYR SARS-CoV-2-M-S10 aa191-221
 SGFAAYSRYRIGNYKLNTDHSSSSDNI ALLV Extended SARS-CoV2-M-E1 aa6-50
 GTITVEELKKLLEQWNLVIGFLFTWIC peptide LLQFAYANRNRFLYIIK SARS-CoV2-M-E2
 aa43-85 NNRNFLYIIKLIFLWLLWPVTLACFVL AVYRINWITGGIAIAMA SARS-CoV2-M-
 E3 aa76-120 ITGGIAIAMACLVGLMWLSYFIASFRLF ARTRSMWSFNPETNILL SARS-CoV2-
 M-E4 aa111-155 SFNPETNILLNVPLHGTILTRP LLESELV IGAVILRGHLRIAGHH SARS-
 CoV2-M-E5 aa146-190 RGHLRIAGHHLGRCDIKDLPKEITVAT SRTL SYYKLGASQRVAGD
 SARS-CoV2-M-E6 aa181-221 LGASQRVAGDSGFAAYSRYRIGNYKL NTDHSSSSDNIALLV
 Short protein SARS-CoV2-M-P1 aa6-116 GTITVEELKKLLEQWNLVIGFLFTWIC
 LLQFAYANRNRFLYIIKLIFLWLLWPV TLACFVLA AVYRINWITGGIAIAMACL
 VGLMWLSYFIASFRLFARTRSMWSFNP ET SARS-CoV2-M-P2 aa106-221
 TRSMWSFNPETNILLNVPLHGTILTRP LLESELVIGAVIL
 RGHLRIAGHHLGRCDIKDLPKEITVAT SRTL SYYKLGASQRVAGDSGFAAYSRY
 RIGNYKLNTDHSSSSDNIALLV Full length SARS-CoV-2-FL aa6-221
 GTITVEELKKLLEQWNLVIGFLFTWIC target LLQFAYANRNRFLYIIKLIFLWLLWPV
 TLACFVLA AVYRINWITGGIAIAMACL VGLMWLSYFIASFRLFARTRSMWSFNP
 ETNILLNVPLHGTILTRP LLESELVIGA VILRGHLRIAGHHLGRCDIKDLPKEIT
 VATSRTL SYYKLGASQRVAGDSGFAA YSRYRIGNYKLNTDHSSSSDNIALLV .sup.1aa
 positions based on SARS-CoV-2 reference genome (NC-06577.2 (Wuhan strain) and reference M
 protein sequences, YP_009724393.1. .sup.2For conjugation to polyionic VLPS, the peptide/ protein
 antigens have a N-terminal TAG and AAYY proteolytic processing sequence.
 TABLE-US-00004 TABLE 4 SARS-CoV-2 antigens of the N structural protein
 Immunogen Seq ID aa-position.sup.1 Amino acid sequence.sup.2 Short SARS-CoV-2-N-
 aa51-84 SWFTALTQH GKEDLKFPRGQGV PINTNSSP peptide S1 DDQI SARS-CoV-2-N- aa75-
 108 NTNSSPDDQIGYYRRATRRIRGGDGKMKD S2 LSPRW SARS-CoV-2-N- aa99-132
 GKMKDLSRWYFY YLGTGPEAGLPYGAN S3 KDGI IW SARS-CoV-2-N- aa123-156
 YGANKDGI IWVATEGALNTPKDHIGTRNP S4 ANNA SARS-CoV-2-N- aa147-180
 GTRNPANNA AIVLQLPQGTTLPKGFYAEG S5 SRGGS SARS-CoV-2-N- aa171-204
 FYAEGSRGGSQASSRSSRSRNSSRNSTPGSS S6 RG SARS-CoV-2-N- aa195-227
 RNSTPGSSRGTS PARMAGNGGDAALALL S7 LDRLN SARS-CoV-2-N- aa219-252
 LALLLLDRLNQLESKMSGKGQQQQGQTV S8 TKKSAA SARS-CoV-2-N- aa243-276
 GQTVTKKSAAEASKKPRQKRTATKAYNVT S9 QAFGR SARS-CoV-2-N- aa267-300
 AYNVTQAFGRRGPEQTQGNFGDQELIRQG S10 TDYKH SARS-CoV-2-N- aa291-324
 LIRQGT DYKHWPQIAQFAPSASAFFGMSRI S11 GMEV SARS-CoV-2-N- aa315-347
 FGMSRIGMEVTPSGTWLT YTGAIKLDDKD S12 PNFK SARS-CoV-2-N- aa338-369
 KLDDKDPNFKDQVILLNKHIDAYKTFPPTE S13 PK Extended SARS-CoV2-N- aa51-104

SWFTALTQHGKEDLKFPARGQGVPIINTNSSP peptide E1
DDQIGYYRRATRRIRGGDGKMKDL SARS-CoV2-N- aa95-148
RGGDGKMKDLSPRWYFYLLGTGPEAGLPY E2 GANKDGIWVATEGALNTPKDHIGT
SARS-CoV2-N- aa139-192 LNTPKDHIGTRNPANNAIIVLQLPQGTTT E3
PKGFYAEGSRGGSQASSRSSSRSRN SARS-CoV2-N- aa183-235
SSRSSSRSRNSSRNSTPGSSRGTSPARMAGN E4 GGDAALALLLLDRLNQLESKMS SARS-
CoV2-N- aa227-280 LNQLESKMSGKGQQQQGQTVTKKSAAEA E5
SKKPRQKRTATKAYNVTQAFGRRGPE SARS-CoV2-N- aa271-325
TQAFGRRGPEQTQGNFGDQELIRQGTDYK E6 HWPQIAQFAPSASAFFGMSRIGMEVT
SARS-CoV2-N- aa316-369 GMSRIGMEVTPSGTWLTYTGAIKLDDDKP E7
NFKDQVILLNKHIDAYKTFPPTEPK Short SARS-CoV2-N- aa51-162
SWFTALTQHGKEDLKFPARGQGVPIINTNSSP protein P1
DDQIGYYRRATRRIRGGDGKMKDLSPRWY FYLLGTGPEAGLPYGANKDGIWVATEGA
LNTPKDHIGTRNPANNAIIVLQLPG SARS-CoV2-N- aa153-265
NNAIIVLQLPQGTTTLPKGFYAEGSRGGSQ P2
ASSRSSSRSRNSSRNSTPGSSRGTSPARMAG NGGDAALALLLLDRLNQLESKMSGKGQQ
QQGQTVTKKSAAEASKKPRQKRTAT SARS-CoV2-N- aa255-369
SKKPRQKRTATKAYNVTQAFGRRGPEQTQ P3 GNFGDQELIRQGTDYKHWPQIAQFAPSAS
AFFGMSRIGMEVTPSGTWLTYTGAIKLDDK DPNFKDQVILLNKHIDAYKTFPPTEPK Full
length SARS-CoV-2-N- aa51-369 SWFTALTQHGKEDLKFPARGQGVPIINTNSSP FL
DDQIGYYRRATRRIRGGDGKMKDLSPRWY (319 aa)
FYLLGTGPEAGLPYGANKDGIWVATEGA LNTPKDHIGTRNPANNAIIVLQLPQGTTT
PKGFYAEGSRGGSQASSRSSSRSRNSSRNST PGSSRGTSPARMAGNGGDAALALLLLDRL
NQLESKMSGKGQQQQGQTVTKKSAAEAS KKPRQKRTATKAYNVTQAFGRRGPEQTQ
GNFGDQELIRQGTDYKHWPQIAQFAPSAS AFFGMSRIGMEVTPSGTWLTYTGAIKLDDK
DPNFKDQVILLNKHIDAYKTFPPTEPK .sup.1aa positions based on SARS-CoV-2 reference
genome (NC-06577.2 (Wuhan strain) and reference N protein sequences, YP_009724397.2.
.sup.2For conjugation to polyionic VLPS, the peptide/protein antigens have a N-terminal TAG and
AAYY proteolytic processing sequence.

TABLE-US-00005 TABLE 5 SARS-CoV-2 antigens of the S2 structural protein.
Immunogen Seq ID aa-position.sup.1 Amino acid sequence.sup.2 Extended SARS-CoV2-
aa671-725 CASYQTQTNSPRRARSVASQSIIAYTMSLGA peptide S2-E1 ENSVAY
SNNSIAIPTNFTISVTTE SARS-CoV2- aa716-769
TNFTISVTTEILPVSMTKTSVDCTMYICGDST S2-E2 ECSNLL LQYGSFCTQLNRALTG
SARS-CoV2- aa760-811 CTQLNRALTGIAVEQDKNTQEVFAQVKQIY S2-E3 KTPPIKDF
GGFNFSQILPDPSKPSK SARS-CoV2- aa805-860
ILPDPSKPSKRSFIEDLLFNKVTLADAGFIKQ S2-E4 YGDCLGDIA ARDLICAQKFNGLTV
SARS-CoV2- aa851-905 CAQKFNGLTVLPPLLTDEMIAQYTSALLAG S2-E5 TITSGWTF
GAGAALQIPFAMQMAYR SARS-CoV2- aa897-951
PFAMQMAYRFNGIGVTQNVLYENQKLIAN S2-E6 QFNSAIG KIQDSLSTASALGKLQDV
SARS-CoV2- aa942-996 ASALGKLQDVVNQNAQALNTLVKQLSSNF S2-E7 GAISSVLN
DILSRLDKVEAEVQIDRL SARS-CoV2- aa987-1041
VEAEVQIDRLITGRLQSLQTYVTQQLIRAAEI S2-E8 RASANLA ATKMSECVLGQSKRVD
SARS-CoV2- aa1032-1087 CVLGQSKRVDFCGKGYHLSFPQSAPHGV S2-E9 VFLHVTYV
PAQEKNTTAPAICHGDKA SARS-CoV2- aa1078-1132
APAICHGDKAHFPREGVFVSNNGTHWFVTQ S2-E10 RNFYEP QIITDNTFVSGNCDVVIGI
SARS-CoV2- aa1123-1177 SGNCDVVIGIVNNTVYDPLQPELDSFKEELD S2-E11 KYFKNHT
SPDVDLGDISGINASVV SARS-CoV2- aa1168-1222
DISGINASVVNIQKEIDRLNEVAKNLNESLI S2-E12 DLQELGKYE QYIKWPWYIWLGFIA
SARS-CoV2- aa1213-1268 PWYIWLGFIAGLIAIVMVTIMLCCMTSCCSC S2-E13

LKCCSCGSCS CCKFDEDDSEPVLKGV Short protein SARS-CoV2- aa671-798
 CASYQTQTNSPRRARSVASQSIIAYTMSLGA S2-P1 ENSVAYSNN
 SIAIPTNFTISVTTEILPVSMTKTSVDCTMYIC GDSTECSNLL
 LQYGSFCTQLNRALTGIAVEQDKNTQEVFA QVKQIYKTPPIKDFG SARS-CoV2- aa788-916
 IYKTPPIKDFGGFNFSQILPDPSKPSKRSFIED S2-P2 LLFNKVTLADA
 GFIKQYGDCLGDIAARDLICAQKFNGLTVL PLLTDEMIAQYTSALLAGTITSGWTFGAG
 AALQIPFAMQMAYRFNGIGVTQNVL SARS-CoV2- aa906-1033
 FNGIGVTQNVLYENQKLIANQFNSAIGKIQ S2-P3 DSLSSTASALGK
 LQDVVNQNAQALNTLVKQLSSNFGAISSVL NDILSRLDKVE
 AEVQIDRLITGRLQSLQTYVTQQLIRAAEIR ASANLAATKMSECV SARS-CoV2- aa1023-
 1152 NLAATKMSECVLGQSKRVDFCGKGYHLMS S2-P4 FPQAPHGVV
 FLHVTYVPAQEKNFTTAPAICHGKAHFPR EGVFVSNG
 THWFVTQRNFYEPQIITDNTFVSGNCDVVI GIVNNTVYD PLQPELIDSFKEEL SARS-
 CoV2- aa1142-1268 QPELIDSFKEELDKYFKNHTSPDVDLGDISGI S2-P5 NASVVNIQKE
 DRLNEVAKNLNESLIDLQELGKYEQYIKWP WYIWLGFIA
 LIAIVMTIMLCCMTSCCCLKGCCSCGSCC KFEDEDDSEPVLKGV .sup.1aa positions based
 on SARS-CoV-2 reference genome (NC-06577.2 (Wuhan strain) and reference spike protein
 sequences, YP_009724390.1. .sup.2For conjugation to polyionic VLPs, the peptide/protein
 antigens have a N-terminal TAG and AAYY proteolytic processing sequence.
 TABLE-US-00006 TABLE 6 SARS-CoV-2 antigens of the ORF3a structural protein aa-
 Immunogen Seq ID position.sup.1 Amino acid sequence.sup.2 Short SARS- aa105-132
 FLYLYALVYFLQSINFVR II peptide CoV2- MRLWLCWK ORF3a- S1 SARS- aa123-147
 IIMRLWLCWKCRSKNPLLYD CoV2- ANYFL ORF3a- S2 SARS- aa138-165
 PLYDANYFLCWHTNCYDYC CoV2- IPYNSVTS ORF3a- S3 SARS- aa156-183
 YCIPYNSVTSSIVITSGDGT CoV2- TSPISEHD ORF3a- S4 SARS- aa174-201
 GTTSPISEHDYQIGGYTEKW CoV2- ESGVKDCV ORF3a- S5 SARS- aa192-219
 KWESGVKDCVVLHSYFTSDY CoV2- YQLYSTQL ORF3a- S6 SARS- aa210-238
 DYYQLYSTQLSTDGTGVEHVT CoV2- FFIYNKIVD ORF3a- S7 SARS- aa229-257
 TFFIYNKIVDEPEEHVQIHT CoV2- IDGSSGVVN ORF3a- S8 SARS- aa248-275
 TIDGSSGVVNPVMEPIYDEP CoV2- TTTTSVPL ORF3a- S9 Extended SARS- aa105-154
 FLYLYALVYFLQSINFVR II peptide CoV2- MRLWLCWKCRSKNPLLYDAN ORF3a-
 YFLCWHTNCY E1 SARS- aa145-194 YFLCWHTNCYDYCIPYNSVT CoV2-
 SSIVITSGDGTTSPISEHDY ORF3a- QIGGYTEKWE E2 SARS- aa185-234
 QIGGYTEKWESGVKDCVVLH CoV2- SYFTSDYYQLYSTQLSTDGT ORF3a- VEHVTFFIYN
 E3 SARS- aa225-275 VEHVTFFIYNKIVDEPEEHV CoV2- QIHTIDGSSGVVNPVMEPIY
 ORF3a- DEPTTTTSVPL E4 Short SARS- aa105-195 FLYLYALVYFLQSINFVR II protein CoV2-
 MRLWLCWKCRSKNPLLYDAN ORF3a- YFLCWHTNCYDYCIPYNSVT P1
 SSIVITSGDGTTSPISEHDY QIGGYTEKWES SARS- aa185-275
 QIGGYTEKWESGVKDCVVLH CoV2- SYFTSDYYQLYSTQLSTDGT ORF3a-
 VEHVTFFIYNKIVDEPEEHV P2 QIHTIDGSSGVVNPVMEPIY DEPTTTTSVPL Full- SAR-
 aa105-275 FLYLYALVYFLQSINFVR II length CoV2- MRLWLCWKCRSKNPLLYDAN ORF3a-
 YFLCWHTNCYDYCIPYNSVT FL SSIVITSGDGTTSPISEHDY (171
 QIGGYTEKWESGVKDCVVLH aa) SYFTSDYYQLYSTQLSTDGT
 VEHVTFFIYNKIVDEPEEHV QIHTIDGSSGVVNPVMEPIY DEPTTTTSVPL .sup.1aa positions
 based on SARS-CoV-2 reference genome (NC-06577.2 (Wuhan strain) and reference ORF3a
 protein sequences, YP_009724391.1. .sup.2For conjugation to polyionic VLPs, the peptide/protein
 antigens have a N-terminal TAG and AAYY proteolytic processing sequence.
 TABLE-US-00007 TABLE 7 SARS-CoV-2 antigens of the ORF7a structural protein aa-
 Immunogen Seq ID position.sup.1 Amino acid sequence.sup.2 Short SARS- aa1-28
 MKIILFLALITLATCELYHY peptide CoV2- QECVRGTT ORF7a- S1 SARS- aa20-47

YQECVRGTTVLLKEPCSSGT CoV2- YEGNSPFH ORF7a- S2 SARS- aa38-65
 GTYEGNSPFHPLADNKFALT CoV2- CFSTQFAF ORF7a- S3 SARS- aa56-84
 LTCFSTQFAFACPDGVKHVY CoV2- QLRARSVSP ORF7a- S4 SARS- aa75-103
 YQLRARSVSPKLFIRQEEVQ CoV2- ELYSPIFLI ORF7a- S5 SARS- aa94-121
 QELYSPIFLIVAAIVFITLC CoV2- FTLKRKTE ORF7a- S6 Extended SARS- aa1-46
 MKIILFLALITLeATCELYH peptide CoV-2- YQECVRGTTVLLKEPCSSGT ORF7a-
 YEGNSPFH E1 SARS- aa38-84 GTYEGNSPFHPLADNKFALT CoV-2-
 CFSTQFAFACPDGVKHVYQL ORF7a- RARSVSP E2 SARS- aa75-121
 YQLRARSVSPKLFIRQEEVQ CoV-2- ELYSPIFLIVAAIVFITLCF ORF7a- TLKRKTE E3 Full
 SARS- aa1-121 MKIILFLALITLATCELYHY length CoV2- QECVRGTTVLLKEPCSSGT
 ORF7a- EGNPFHPLADNKFALTCFS FL TQFAFACPDGVKHVYQLRAR
 SVSPKLFIRQEEVQELYSPI FLIVAAIVFITLCFTLKRKT E .sup.1aa positions based on SARS-
 CoV-2 reference genome (NC-06577.2 (Wuhan strain) and reference ORF7a protein sequences,
 YP_009724395.1 .sup.2For conjugation to polyionic VLPs, the peptide/protein antigens have a N-
 terminal TAG and AAYY proteolytic processing sequence.
 TABLE-US-00008 TABLE 8 SARS-CoV-2 antigens of nsp6 and nsp7 nonstructural
 proteins aa- Immunogen Seq ID position.sup.1 Amino acid sequence.sup.2 Short SARS- 1-
 30 SAVKRTIKGTHHWLLTILT peptide CoV-2 SLLVLVQSTQ nsp6-S1 SARS- 21-50
 SLLVLVQSTQWSLFFFLYEN CoV-2 AFLPFAMGII nsp6-S2 SARS- 41-70
 AFLPFAMGIIAMSAFAMMFV CoV-2 KHKHAFLCLF nsp6-S3 SARS- 61-90
 KHKHAFLCLFLLPSLATVAY CoV-2 FNMVYMPASW nsp6-S4 SARS- 81-110
 FNMVYMPASWVMRIMTWLDM CoV-2 VDTSLSGFKL nsp6-S5 SARS- 101-130
 VDTSLSGFKLKDCVMYASAV CoV-2 VLLILMTART nsp6-S6 SARS- 121-150
 VLLILMTARTVYDDGARRVW CoV-2 TLMNVLTLYVY nsp6-S7 SARS- 141-170
 TLMNVLTLYVYKVYYGNALDQ CoV-2 AISMWALIIS nsp6-S8 SARS- 161-190
 AISMWALIISVTSNYSGVVT CoV-2 TVMFLARGIV nsp6-S9 SARS- 181-210
 TVMFLARGIVFMCVEYCPIF CoV-2 FITGNTLQCI nsp6-S10 SARS- 201-230
 FITGNTLQCIMLVYCFLGYF CoV-2 CTCYFGLFCL nsp6-S11 SARS- 221-250
 CTCYFGLFCLLNRYFRLTLG CoV-2 VYDYLVTSTQE nsp6-S12 SARS- 241-270
 VYDYLVTSTQEFYRMNSQGLL CoV-2 PPKNSIDAFK nsp6-S13 SARS- 261-290
 PPKNSIDAFKLNKLLGVGG CoV-2 KPCIKVATVQ nsp6-S14 SARS- 1-28
 SKMSDVKCTSVVLLSVLQQL CoV-2 RVESSSKL nsp7-S1 SARS- 19-46
 QLRVESSSKLWAQCVQLHND CoV-2 ILLAKDTT nsp7-S2 SARS- 37-64
 NDILLAKDTTEAFEKMSVLL CoV-2 SVLLSMQG nsp7-S3 SARS- 55-83
 LLSVLLSMQGAVDINKLCEE CoV-2 MLDNRATLQ nsp7-S4 Extended SARS- 1-50
 SAVKRTIKGTHHWLLTILT peptide CoV-2 SLLVLVQSTQWSLFFFLYEN nsp6-E1
 AFLPFAMGII SARS- 41-90 AFLPFAMGIIAMSAFAMMFV CoV-2
 KHKHAFLCLFLLPSLATVAY nsp6-E2 FNMVYMPASW SARS- 81-130
 FNMVYMPASWVMRIMTWLDM CoV-2 VDTSLSGFKLKDCVMYASAV nsp6-E3
 VLLILMTART SARS- 121-170 VLLILMTARTVYDDGARRVW CoV-2
 TLMNVLTLYVYKVYYGNALDQ nsp6-E4 AISMWALIIS SARS- 161-210
 AISMWALIISVTSNYSGVVT CoV-2 TVMFLARGIVFMCVEYCPIF nsp6-E5 FITGNTLQCI
 SARS- 201-250 FITGNTLQCIMLVYCFLGYF CoV-2 CTCYFGLFCLLNRYFRLTLG nsp6-E6
 VYDYLVTSTQE SARS- 241-290 VYDYLVTSTQEFYRMNSQGLL CoV-2
 PPKNSIDAFKLNKLLGVGG nsp6-E7 KPCIKVATVQ SARS- 1-46
 SKMSDVKCTSVVLLSVLQQL CoV-2 RVESSSKLWAQCVQLHNDIL nsp7-E1 LAKDTT
 SARS- 37-83 NDILLAKDTTEAFEKMSVLL CoV-2 SVLLSMQGAVDINKLCEEML nsp7-E2
 DNRATLQ Short SARS- 1-104 SAVKRTIKGTHHWLLTILT protein CoV-2
 SLLVLVQSTQWSLFFFLYEN nsp6-SP1 AFLPFAMGIIAMSAFAMMFV
 KHKHAFLCLFLLPSLATVAY FNMVYMPASWVMRIMTWLDM VDTSL SARS- 94-197

IMTWLDMVTLNMTWLDL CoV-2 VMYASAVVLLILMTARTVYD nsp6-SP2
 DGARRVWTLMNVLTLVYKVV YGNALDQAISMWALIISVTS
 NYSGVVTTVMFLARGIVFMC VEYC SARS- 187-290 RGIVFMCVEYCPITFITGNT CoV-2
 LQCIMLVYCFLGYFCTCYFG nsp6-SP3 LFCLLNRYFRLTLGVYDYL
 STQEFYRMNSQGLLPPKNSI DAFKLNKLLGVGGKPCIKV ATVQ Full SARS- 1-290
 SAVKRTIKGTHHWLTLT length CoV-2 SLLVLVQSTQWSLFFFLYEN nsp6-FL
 AFLPFAMGIIAMSAFAMMFV KHKHAFLLCLFLLPSLATVAY
 FNMVYMPASWVMRIMTWLDM VDTSLSGFKLKDCVMYASAV
 VLLILMTARTVYDDGARRVW TLMNVLTTLVYKVYYGNALDQ
 AISMWALIISVTSNYSGVVT TVMFLARGIVFMCVEYCPIT FITGNTLQCIMLVYCFLGYF
 CTCYFGLFCLLNRYFRLTLG VYDYLVSSTQEFYRMNSQGLL PPKNSIDAFKLNKLLGVGG
 KPCIKVATVQ SARS- 1-83 SKMSDVKCTSVLLSVLQQL CoV-2
 RVESSSKLWAQCVQLHNDIL nsp7-FL LAKDTTEAFEKMSVLLSVLL
 SMQGAVDINKLCEEMLDNRA TLQ .sup.1aa positions based on SARS-CoV-2 reference genome
 (NC-06577.2; Wuhan strain) and reference nsp6 protein (YP_009725302.1) and nsp7 protein
 (YP_009725303.1). .sup.2For conjugation to polyionic VLPS, the peptide/protein antigens have a
 N-terminal TAG and AAYY proteolytic processing sequence.

TABLE-US-00009 TABLE 9 SARS-CoV-2 antigens of nsp12-1 nonstructural protein
 fragment aa- Immunogen Seq ID position.sup.1 Amino acid sequence.sup.2 Short SARS-
 126-153 DLVYALRHFDEGNCDTLKEI peptide CoV-2 LVTYNCCD nsp12- 1-S1 SARS- 144-
 171 EILVTYNCCDDDDYFNKKDWY CoV-2 DFVENPDI nsp12- 1-S2 SARS- 162-189
 WYDFVENPDILRVYANLGER CoV-2 VRQALLKT nsp12- 1-S3 SARS- 180-207
 ERVRQALLKTVQFCDAMRNA CoV-2 GIVGVLT nsp12- 1-S4 SARS- 198-225
 NAGIVGVLTLDNQDLNGNWY CoV-2 DFGDFIQT nsp12- 1-S5 SARS- 216-243
 WYDFGDFIQTTPGSGVPVVD CoV-2 SYYSLLMP nsp12- 1-S6 SARS- 234-261
 VDSYYSLLMPILTLTRALTA CoV-2 ESHVDTDL nsp12- 1-S7 SARS- 252-280
 TAESHVDTDLTKPYIKWDL CoV-2 KYDFTEERL nsp12- 1-S8 SARS- 271-299
 LKYDFTEERLKLFDYFKYW CoV-2 DQTYHPNCV nsp12- 1-S9 SARS- 290-317
 WDQTYHPNCVNCLDDRCILH CoV-2 CANFNVLF nsp12- 1-S10 SARS- 308-336
 LHCANFNVLFSTVFPPTSFG CoV-2 PLVRKIFVD nsp12- 1-S11 SARS- 327-355
 GPLVRKIFVDGVPFVVSTGY CoV-2 HFRELGVVH nsp12- 1-S12 SARS- 346-375
 YHFRELGVVHNQDVNLHSSR CoV-2 LSFKELLVYA nsp12- 1-S13 Extended SARS- 126-175
 DLVYALRHFDEGNCDTLKEI peptide CoV-2 LVTYNCCDDDDYFNKKDWYDF nsp12-
 VENPDILRVY 1-E1 SARS- 166-215 VENPDILRVYANLGERVRQA CoV-2
 LLKTVQFCDAMRNAGIVGVLTLDNQDLNGN 1-E2 SARS- 206-255
 TLDNQDLNGNWYDFGDFIQT CoV-2 TPGSGVPVVDVSYYSLLMPIL nsp12- TLTRALTAES 1-
 E3 SARS- 246-295 TLTRALTAESHVDTDLTKPY CoV-2 IKWDLCLKYDFTEERLKLFD nsp12-
 YFKYWDQTYH 1-E4 SARS- 286-335 YFKYWDQTYHPNCVNCLDDR CoV-2
 CILHCANFNVLFSTVFPPTS nsp12- FGPLVRKIFV 1-E5 SARS- 326-375
 FGPLVRKIFVDGVPFVVSTG CoV-2 YHFRELGVVHNQDVNLHSSR nsp12- LSFKELLVYA 1-
 E6 Short SARS- 126-255 DLVYALRHFDEGNCDTLKEI protein CoV-2
 LVTYNCCDDDDYFNKKDWYDF nsp12- VENPDILRVYANLGERVRQA 1-SP1
 LLKTVQFCDAMRNAGIVGVLTLDNQDLNGNWYDFGDFIQT
 TPGSGVPVVDVSYYSLLMPIL TLTRALTAES SARS- 246-375 TLTRALTAESHVDTDLTKPY
 CoV-2 IKWDLCLKYDFTEERLKLFD nsp12- YFKYWDQTYHPNCVNCLDDR 1-SP2
 CILHCANFNVLFSTVFPPTS FGPLVRKIFVDGVPFVVSTG YHFRELGVVHNQDVNLHSSR
 LSFKELLVYA Full SARS- 126-375 DLVYALRHFDEGNCDTLKEI length CoV-2
 LVTYNCCDDDDYFNKKDWYDF antigen nsp12- VENPDILRVYANLGERVRQA 1-FL
 LLKTVQFCDAMRNAGIVGVLTLDNQDLNGNWYDFGDFIQT
 TPGSGVPVVDVSYYSLLMPIL TLTRALTAESHVDTDLTKPY IKWDLCLKYDFTEERLKLFD

YFKYWDQYTLPCNVCLDDR CILHACNFVNLFLSTVFPPTS FGPLVRKIFVDGVPFVSTG
YHFRELGVVHNQDVNLHSSR LSFKELLVYA .sup.1aa positions based on SARS-CoV-2
reference genome (NC-06577.2; Wuhan strain) and reference nsp12 protein (YP_009725307.1).
.sup.2For conjugation to polyionic VLPS, the peptide/protein antigens have a N-terminal TAG and
AAYY proteolytic processing sequence.

TABLE-US-00010 TABLE 10 SARS-CoV-2 antigens of nsp12-2 nonstructural protein
fragment aa- Immunogen Seq ID position Amino acid sequence Short SARS- 520-548
SYEDQDALFAYTKRNV IPT peptide CoV-2 ITQMNLKYAI nsp12- 2-S1 SARS- 538-566
TITQMNLKYAISAKNRART CoV-2 VAGVSICSTM nsp12- 2-S2 SARS- 557-586
VAGVSICSTMNRQFHQKLL CoV-2 KSIAATRGAT nsp12- 2-S3 SARS- 576-604
LKSIAATRGATVVIGTSKF CoV-2 YGGWHNMLKT nsp12- 2-S4 SARS- 595-624
YGGWHNMLKTVYSDVENPH CoV-2 LMGWDYPKCDR nsp12- 2-S5 SARS- 615-643
MGWDYPKCDRAMPNMLRIM CoV-2 ASLVLARKHT nsp12- 2-S6 SARS- 634-662
ASLVLARKHTTCCSLSHRF CoV-2 YRLANECAQV nsp12- 2-S7 SARS- 653-683
YRLANECAQVLSEMVMCGG CoV-2 SLYVKPGGTSSG nsp12- 2-S8 SARS- 674-704
YVKPGGTSSGDATTAYANS CoV-2 VFNICQAVTANV nsp12- 2-S9 SARS- 695-724
NICQAVTANVNALLSTDGN CoV-2 KIADKYVRNLQ nsp12- 2-S10 SARS- 715-744
IADKYVRNLQHRLYECLYR CoV-2 NRDVD TDFVNE nsp12- 2-S11 SARS- 735-764
RDVD TDFVNEFYAYLRKH F CoV-2 SMMILSDDAVV nsp12- 2-S12 SARS- 755-784
MMILSDDAVVCFNSTYASQ CoV-2 GLVASIKNFKS nsp12- 2-S13 SARS- 775-804
LVASIKNFKSVLYYQNNVF CoV-2 MSEAKCWTETD nsp12- 2-S14 SARS- 795-824
SEAKCWTETDLTKGPHEFC CoV-2 SQHTMLVKQGD nsp12- 2-S15 SARS- 815-844
QHTMLVKQGDDYVYLPYPD CoV-2 PSRILGAGCFV nsp12- 2-S16 SARS- 835-863
SRILGAGCFVDDIVKTDGT CoV-2 LMIERFVSLA nsp12- 2-S17 SARS- 854-882
LMIERFVSLAIDAYPLTKH CoV-2 PNQEYADV FH nsp12- 2-S18 SARS- 873-901
PNQEYADV FHLYLQYIRKL CoV-2 HDELTGHMLD nsp12- 2-S19 SARS- 892-920
HDELTGHMLDMYSVMLTND CoV-2 NTSRYWEPEF nsp12- 2-S20 Extended SARS- 520-498
SYEDQDALFAYTKRNV IPT peptide CoV-2 ITQMNLKYAISAKNRARTV nsp12-
AGVSICSTMTN 2-E1 SARS- 559-607 GVSICSTMTNRQFHQKLLK CoV-2
SIAATRGATVVIGTSKFY G nsp12- GWHNMLKTVYS 2-E2 SARS- 598-646
WHNMLKTVYSDVENPHLMG CoV-2 WDYPKCDRAMPNMLRIMAS nsp12-
LVLARKHTTCC 2-E3 SARS- 637-685 VLARKHTTCCSLSHRFYRL CoV-2
ANECAQVLSEMVMCGGSLY nsp12- VKPGGTSSGDA 2-E4 SARS- 676-724
KPGGTSSGDATTAYANSVF CoV-2 NICQAVTANVNALLSTDGN nsp12- KIADKYVRNLQ 2-
E5 SARS- 715-763 IADKYVRNLQHRLYECLYR CoV-2 NRDVD TDFVNEFYAYLRKH nsp12-
FSMMILSDDAV 2-E6 SARS- 754-802 SMMILSDDAVVCFNSTYAS CoV-2
QGLVASIKNFKSVLYYQNN nsp12- VFMSEAKCWTE 2-E7 SARS- 793-841
FMSEAKCWTETDLTKGPHE CoV-2 FCSQHTMLVKQGDDYVYLP nsp12- YPDPSRILGAG 2-
E8 SARS- 832-880 PDPSRILGAGCFVDDIVKT CoV-2 DGTLMIERFVSLAIDAYPL nsp12-
TKHPNQEYADV 2-E9 SARS- 871-920 KHPNQEYADV FHLYLQYIR CoV-2
KLHDELTGHMLDMYSVMLT nsp12- NDNTSRYWEPEF 2-E10 Short SARS- 520-629
SYEDQDALFAYTKRNV IPT protein CoV-2 ITQMNLKYAISAKNRARTV nsp12-
AGVSICSTMTNRQFHQKLL 2-SP1 KSIAATRGATVVIGTSKFY
GGWHNMLKTVYSDVENPHL MGWDYPKCDRAMPNM SARS- 619-727
YPKCDRAMPNMLRIMASLV CoV-2 LARKHTTCCSLSHRFYRLA nsp12-
NECAQVLSEMVMCGGSLYV 2-SP2 KPGGTSSGDATTAYANSVF
NICQAVTANVNALLSTDGN KIADKYVRNLQHRL SARS- 717-826
DKYVRNLQHRLYECLYRNR CoV-2 DVD TDFVNEFYAYLRKHFS nsp12-
MMILSDDAVVCFNSTYASQ 2-SP3 GLVASIKNFKSVLYYQNNV
FMSEAKCWTETDLTKGPHE FCSQHTMLVKQGDDY SARS- 816-920

HTMLVQGDYDPDP CoV-2 SRILGAGCFVDDIVKTDGT nsp12-
 LMIERFVSLAIDAYPLTKH 2-SP4 PNQEYADV FHLYLQYIRKL
 HDELTGHMLDMYSVMLTND NTSRYWEPEF Full SARS- 520-920
 SYEDQDALFAYTKRNV IPT length CoV-2 ITQMNLKYAISAKNRARTV antigen nsp12-
 AGVSICSTMTNRQFHQKLL 2-FL KSIAATR GATV VIGTSKFY
 GGWHNMLKTVYSDVENPHL MGWDYPKCDRAMPNMLRIM ASLVLARKHTTCCSLSHRF
 YRLANCAQVLSEVMVCGG SLYVKPGGTSSGDATTAYA NSVFNICQAVTANVNALLS
 TDGNKIADKYVRNLQHRLY ECLYRNRD VDTDFVNEFYA YLRKHFSMMILSDDAVVCF
 NSTYASQGLVASIKNFKSV LYYQNNVFMSEAKCWTETD LTKGPHEFCSQHTMLVKQG
 DDYVYLPYPDPSRILGAGC FVDDIVKTDGTLMIERFVS LAIDAYPLTKHPNQEYADV
 FHLYLQYIRKLHDELTGHM LDMYSVMLTNDNTSRYWEP EF .sup.1aa positions based on
 SARS-CoV-2 reference genome (NC-06577.2; Wuhan strain) and reference nsp12 protein
 (YP_009725307.1). .sup.2For conjugation to polyionic VLPS, the peptide/protein antigens have a
 N-terminal TAG and AAYY proteolytic processing sequence.

TABLE-US-00011 TABLE 11 OC43 and HKU1 antigens of the M structural protein
 aa- Immunogen Seq ID position.sup.1 Amino acid sequence.sup.2 Short OC43-M-S1 aa14-42
 TADEAIKFLKEWNFSLGIIL peptides LFITIILQF OC43-M-S2 aa33-60
 LLFITIILQFGYTSRSMFV YVIKMILW OC43-M-S3 aa51-79 VYVIKMILWLMWPLTIILT
 IFNCVYALN OC43-M-S4 aa70-98 TIFNCVYALNNVYLGLSIVF TIVAIIMWI OC43-M-S5
 aa89-117 FTIVAIIMWIVYFVNSIRLF IRTGSFWSF OC43-M-S6 aa108-136
 FIRTGSFWSFNPETNNLMCI DMKGTMYVR OC43-M-S7 aa127-154
 IDMKGTMYVRPIIEDYHTLT VTIIRGHL OC43-M-S8 aa145-172 LTVTIIRGHLYIQGIKLTG
 YSLADLPA OC43-M-S9 aa163-190 TGYSLADLPAYMTVAKVTHL CTYKRGFL OC43-M-
 aa181-208 HLCTYKRGFLDRISDTSGFA S10 VYVKS KVG OC43-M- aa199-226
 FAVYVKS KVGNYRLPSTQKG S11 SGMDTALL HKU1-M-S1 aa75-97
 NNAFLAFSIVFTIISIVIWI LYF HKU1-M-S2 aa174-200 KVQVLCTYKRAFLDKLDVNS
 GFAVFVK Extended OC43-M-E1 aa14-57 TADEAIKFLKEWNFSLGIIL peptides
 LFITIILQFGYTSRSMFVYV IKMI OC43-M-E2 aa48-91 SMFVYVIKMILWLMWPLTI
 ILTIFNCVYALNNVYLGLSI VFTI OC43-M-E3 aa82-125 YLGLSIVFTIVAIIMWIVYF
 VNSIRLFIRTGSFWSFNPET NNLM OC43-M-E4 aa116-159 SFNPETNNLMCIDMKGTMFV
 RPIIEDYHTLT VTIIRGHLY IQGI OC43-M-E5 aa150-194 IRGHLYIQGIKLTGYSLAD
 LPAYMTVAKV TYLCTYKRGF LDKIS OC43-M-E6 aa185-226 YKRGFLDKISDTSGFAVYVK
 SKVGNYRLPSTQKGSGMDTA LL HKU1-M-E1 aa75-97/ NNAFLAFSIVFTIISIVIWI 174-200
 LYF/KVQVLCTYKRAFLDKL DVNSGFAVFVK Short OC43-M-P1 aa14-151
 TADEAIKFLKEWNFSLGIIL proteins LFITIILQFGYTSRSMFVYV IKMIILWLMWPLTIILTIFN
 CVYALNNVYLGLSIVFTIVA IIMWIVYFVNSIRLFIRTGS FWSFNPETNNLMCIDMKGTM
 YVRPIIEDYHTLT VTIIR OC43// aa141-226// DYHTLT VTIIRGHLYIQGIK HKU1- aa75-97/
 LGTGYSLADLPAYMTVAKVT M-P2 aa174-200 HLCTYKRGFLDRISDTSGFA
 VYVKS KVGNYRLPSTQKGSG MDTALL/NNAFLAFSIVFTI ISIVIWILYF/KVQVLCTYK
 RAFLDKLDVNSGFAVFVK Full OC43-M// aa14-226// TADEAIKFLKEWNFSLGIIL length
 HKU1-M- aa75-97/ LFITIILQFGYTSRSMFVYV variant- aa174-200
 IKMIILWLMWPLTIILTIFN FL CVYALNNVYLGLSIVFTIVA (263 aa)
 IIMWIVYFVNSIRLFIRTGS FWSFNPETNNLMCIDMKGTM YVRPIIEDYHTLT VTIIRGH
 LYIQGIKLTGYSLADLPAY MTVAKVTHLCTYKRGFLDRI SDTSGFAVYVKS KVGNYRLP
 STQKGSGMDTALL//NNAFL AFSIVFTIISIVIWILYF/K VQVLCTYKRAFLDKLDVNSG
 FAVFVK .sup.1aa positions based on OC43 reference M protein sequences, YP_009555244.1, and
 HKU1 reference M protein sequence, YP_173241.1. .sup.2For conjugation to polyionic VLPS, the
 peptide/protein antigens have a N-terminal TAG and AAYY proteolytic processing sequence. /
 indicates break in native amino acid sequence // indicates break between OC43 amino acid
 sequences and HKU1 amino acid sequences.

TABLE-US-00012 TABLE 12 OC43 and HKU1 antigens of N structural protein aa
Immunogen Seq ID positions.sup.1 Amino acid sequence.sup.2 Short OC43-N- aa64-88
SWFSGITQFQKGKEFEFVEGQGVPI peptides S1 OC43-N- aa99-131
GYWYRHNRRSFKTADGNQRQLLPRWYF S2 YYLGTG OC43-N- aa122-155
RWYFYLLGTGPHAKDQYGDIDGVYWVA S3 SNQAVD OC43-N- aa146-177
YWVASNQADVNTPADIVDRDPSSDEAIP S4 TRFP OC43-N- aa168-200
SDEAIPTRFPPGTVLPQGYYIEGSGRSAPN S5 SRS OC43-N- aa191-223
SGRSAPNSRSTSRSTSSRASSAGSRSRANS S6 NRT OC43-N- aa214-245
RSRANSNGNRTPTSGVTPDMADQIASLV S7 AKLG OC43-N- aa236-265
IASLVLA KL GKDATKPQQVTKHTAKEVR S8 QK OC43-N- aa270-302
PRQKRSPNKQCTVQQCFGKRGPNQNFG S9 GGEMLK OC43-N- aa293-325
QNFGGGEMLKLGTS DPQFPILAE LAPT A S10 GAFF OC43-N- aa316-340/
LAPTAGAFFFFGSRLELAKVQNL SGN/ELR S11 aa350-358 YNGAIR OC43-N- aa350-381
ELRYNGAIRFDSTLSGFETIMKVLNENLN S12 AYQ HKU1-N- aa131-156
PYANASYGESLEGVFWVANHQADTST S1 HKU1-N- aa147-170
VANHQADTSTPSDVSSRDPTTQEA S2 HKU1-N- aa258-289
RPGSRSQSRGPNNRSLRSNSNFRHSDSIV S3 KP HKU1-N- aa325-341
SKLDLVKRDSEADSPVK S4 Extended OC43-N-1 aa64-88/
SWFSGITQFQKGKEFEFVEGQGVPI/ peptides aa99-119 GYWYRHNRRSFKTADGNQRQL
OC43-N-2 aa110-155 KTADGNQRQLLPRWYFYLLGTGPHAKD
QYGT DIDGVYWVASNQADV OC43-N-3 aa146-191
YWVASNQADVNTPADIVDRDPSSDEAIP TRFPPGTVLPQGYYIEGS OC43-N-4 aa182-224
LPQGYYIEGSGRSAPNSRSTSRSTSSRAS SAGSRSRANSNGNRTPT OC43-N-5 aa215-260
SRANSNGNRTPTSGVTPDMADQIASLVLA KL GKDATKPQQVTKHTAK OC43-N-6 aa251-265/
PQQVTKHTAKEVRQK/ aa270-300 PRQKRSPNKQCTVQQCFGKRGPNQNFG GGEM OC43-
N-7 aa291-336 PNQNFGGGEMLKLGTS DPQFPILAE LAP TAGAFFFFGSRLELAKVQN OC43-
N-8 aa327-340/ SRLELAKVQNL SGN/ELRYNGAIRFDSTL aa350-381
SGFETIMKVLNENLNAYQ KHU1-N- aa131-170 PYANASYGESLEGVFWVANHQADTSTPS
1 DVSSRDPTTQEA KHU1-N- aa198-229/ RPGSRSQSRGPNNRSLRSNSNFRHSDSIV 2
aa325-341 KP/SKLDLVKRDSEADSPVK Short OC43-N- aa64-88/
SWFSGITQFQKGKEFEFVEGQGVPI/ protein P1 aa99-184
GYWYRHNRRSFKTADGNQRQLLPRWYF YYLGTGPHAKDQYGT
DIDGVYWVASNQADVNTPADIVDRDPS SDEAIPTRFPPGTVLPQ OC43-N- aa174-265/
TRFPPGTVLPQGYYIEGSGRSAPNSRSTSR P2 270-283
TSSRASSAGSRSRANSNGNRTPTSGVTPDM ADQIASLVLA KL GKDATKPQQVTKHTA
KEVRQK/PRQKRSPNKQCTVQ OC43-N- aa273-381
KRSPNKQCTVQQCFGKRGPNQNFGGGE P3 MLKGTSDPQFPILAE LAPT AGAFFFFGSRL
ELAKVQNL SGNELRYNGAIRFDSTLSGFE TIMKVLNENLNAYQ HKU-N- aa131-170/
PYANASYGESLEGVFWVANHQADTSTPS P1 198-229/ DVSSRDPTTQEA/ 325-341
RPGSRSQSRGPNNRSLRSNSNFRHSDSIV KP/SKLDLVKRDSEADSPVK Full OC43- aa64-
88/ SWFSGITQFQKGKEFEFVEGQGVPI/ length N/HKU1- aa99-265/
GYWYRHNRRSFKTADGNQRQLLPRWYF N-variant- aa270-381//
YYLGTGPHAKDQYGDIDGVYWVASNQA FL aa131-170/
DVNTPADIVDRDPSSDEAIPTRFPPGTVL (383 aa) 198-229/
PQGYYIEGSGRSAPNSRSTSRSTSSRASSAG 325-341
SRSRANSNGNRTPTSGVTPDMADQIASLV LAKLGKDATKPQQVTKHTAKEVRQK/
PRQKRSPNKQCTVQQCFGKRGPNQNFG GGEMLKLGTS DPQFPILAE LAPT AGAFFFF
GSRLELAKVQNL SGNELRYNGAIRFDSTL SGFETIMKVLNENLNAYQ//
PYANASYGESLEGVFWVANHQADTSTPS DVSSRDPTTQEA/
RPGSRSQSRGPNNRSLRSNSNFRHSDSIV KP/SKLDLVKRDSEADSPVK .sup.1aa positions

based on OC43 reference N protein sequences, YP_009555245.1, and HKU1 reference N protein sequence, YP_173242.1. .sup.2For conjugation to polyionic VLPS, the peptide/protein antigens have a N-terminal TAG and AAYY proteolytic processing sequence. / indicates break in native amino acid sequence // indicates break between OC43 amino acid sequences and HKU1 amino acid sequences.

TABLE-US-00013 TABLE 13 OC43 and HKU1 S2 antigens of the S2 structural protein aa Immunogen Seq ID positions.sup.1 Amino acid sequence.sup.2 Short
OC43-S2- aa898-929 SKASSRSAIEDLLFDKVKLSDVGFVEAY peptide S1 NNCT OC43-S2-
aa920-951 GFVEAYNNCTGGAEIRDLCVQSYKGI S2 KVLPP OC43-S2- aa943-973
SYKGIKVLPPLLSENQISGYTLAATSASL S3 FPP OC43-S2- aa964-995
AATSASLFPWTAAGVPFYLNVQYRI S4 NGLGV OC43-S2- aa986-1017
VQYRINGLGVTMDVLSQNQKLIANAF S5 NNALYA OC43-S2- aa1008-1039
ANAFNNALYAIQEGFDATNSALVKIQ S6 AVVNAN OC43-S2- aa1030-1061
VKIQAVVNANAEALNNLLQQLSNRFG S7 AISASL OC43-S2- aa1052-1084
NRFGAISASLQEILSRDALEAEAQIDRL S8 INGR OC43-S2- aa1075-1107
AQIDRLINGRLTALNAYVSQQLSDSTLV S9 KFSAA OC43-S2- aa1098-1130
DSTLVKFSAAQAMEKVNCEVKSQSSRI S10 NFCGNG OC43-S2- aa1120-1153
QSSRINFCGNGNHI S11 ISLVQNAPYGLYFIHFSYVP/ OC43-S2- aa1228-1258
PNLPDFKEELDQWFKNQTSVAPDLSLD S12 YINVT OC43-S2- aa1250-1281
DLSLDYINVTFLDLQVEMNRLQEAIKV S13 LNQSY OC43-S2- aa1272-1302
EAIKVLNQSINLKDITYEYYVKWPW S14 YVWL HKU1-S2- aa1229-1258
PKLSDFESELHWFKNQTSIAPNLTLNL S1 HT HKU1-S2- aa1249-1280
APNLTLNLHTINATFLDLYYEMNLIQES S2 IKSL Extended OC43-S2- aa898-942
SKASSRSAIEDLLFDKVKLSDVGFVEAY peptide E1 NNCTGGAEIRDLCVQS OC43-S2-
aa933-978 EIRDLCVQSYKGIKVLPPLLSENQISGY E2 TLAATSASLFPWTAAGV OC43-S2-
aa969-1013 SLFPWTAAGVPFYLNVQYRINGLGVTMDVLSQNQKLIANAFNN OC43-
S2- aa1004-1048 QKLIANAFNNALYAIQEGFDATNSALV E4 KIQAVVNANAEALNNLLQ
OC43-S2- aa1039-1083 NAEALNNLLQQLSNRFGAISASLQEILS E5
RLDALEAEAQIDRLING OC43-S2- aa1074-1119 EAQIDRLINGRLTALNAYVSQQLSDSTL E6
VKFSAAQAMEKVNCEVKS OC43-S2- aa1109-1153 AMEKVNCEVKSQSSRINFCGNGNHIIS
E7 LVQNAPYGLYFIHFSYVP/ OC43-S2- aa1228-1269
PNLPDFKEELDQWFKNQTSVAPDLSLD E8 YINVTFLDLQVEMNR OC43-S2- aa1260-1302
FLDLQVEMNRLQEAIKVLNQSINLKD E9 IGTIEYYVKWPWYVWL HKU1-S2- aa1229-
1280 PKLSDFESELHWFKNQTSIAPNLTLNL E1 HTINATFL MNLIQESIKSL Short OC43-
S2- aa898-1033 SKASSRSAIEDLLFDKVKLSDVGFVEAY protein P1
NNCTGGAEIRDLCVQSYKGIKVLPPLL SENQISGYTLAATSASLFPWTAAGV
PFYLNVQYRINGLGVTMDVLSQNQKLI ANAFNNALYAIQEGFDATNSALVKIQ OC43-S2-
aa1022-1153 FDATNSALVKIQAVVNANAEALNNLL P2
QQLSNRFGAISASLQEILSRDALEAEA QIDRLINGRLTALNAYVSQQLSDSTLVK
FSAAQAMEKVNCEVKSQSSRINFCGNG NHIISLVQNAPYGLYFIHFSYVP/ OC43/HK
aa1228-1302// PNLPDFKEELDQWFKNQTSVAPDLSLDY U1- 1229-1280
INVFLDLQVEMNRLQEAIKVLNQSIN variant-S2-
LKDITYEYYVKWPWYVWL//PKLSDF P3 ESELHWFKNQTSIAPNLTLNLHTINAT
FLMNLIQESIKSL .sup.1aa positions based on OC43 reference Spike protein sequences,
YP_009555241.1, and HKU1 reference spike protein sequence, YP_173238.1. .sup.2For
conjugation to polyionic VLPS, the peptide/protein antigens have a N-terminal TAG and AAYY
proteolytic processing sequence. / indicates break in native amino acid sequence // indicates break
between OC43 amino acid sequences and HKU1 amino acid sequences

TABLE-US-00014 TABLE 14 OC43 antigens of nonstructural proteins nsp3 and
nsp4 Immunogen Seq ID aa positions.sup.1 Amino acid sequence.sup.2 Short OC43-

aa2368-2399 FMRFYIIIASFIFSLFRHVAYGCSKSGCLF peptide nsp3-S1 OC43- aa2390-2422
YGCSKSGCLFYKRNRLRVKLCSTIVGGMIRYY nsp3-S2 OC43- aa2413-2442
TIVGGMIRYYDVMANGGTGFCSKHQWNCID nsp3-S3 OC43- aa2433-2463
CSKHQWNCIDCDSYKPGNTFITVEAALDLSK nsp3-S4 OC43- aa2454-2473/
TVEAALDLSKELKRPIQPTD/NAAVFYAQLSLFR nsp3-S5 aa2542-2553 OC43- aa2544-2574
AVFYAQLSLFRPILMVDKNLITTANTGTSVTE nsp3-S6 OC43- aa2565-2595
TANTGTSVTETMFDVYVDTFSLMFDVDKKSL nsp3-S7 OC43- aa2586-2602/
SMFDVDKKSLNALIATA/ELTDESCNNLVPTYL nsp3-S8 aa2651-2665 OC43- aa2656-2686
SCNNLVPTYLKSDNIVAADLGVLIQNSAKHV nsp3-S9 OC43- aa2677-2705
VLIQNSAKHVQGNVAKIAGVSCIWSVDAF nsp3-S10 Extended OC43- aa2368-2409
FMRFYIIIASFIFSLFRHVAYGCSKSGCLFCYKRN peptide nsp3-E1 RSLRV OC43- aa2400-
2442 CYRNRSLRVKLCSTIVGGMIRYYDVMANGGTGFCSK nsp3-E2 HQWNCID OC43-
aa2433-2473 CSKHQWNCIDCDSYKPGNTFITVEAALDLSKELKR nsp3-E3 PIQPTD/ OC43-
aa2542-2585 NAAVFYAQLSLFRPILMVDKNLITTANTGTSVTETM nsp3-E4 FDVYVDTF
OC43- aa2576-2602/ MFDVYVDTFSLMFDVDKKSLNALIATA/ nsp3-E5 aa2651-2673
ELTDESCNNLVPTYLKSDNIVA OC43- aa2664-2705
YLKSDNIVAADLGVLIQNSAKHVQGNVAKIAGVS nsp3-E6 CIWSVDAF Short OC43-
aa2368-2473 FMRFYIIIASFIFSLFRHVAYGCSKSGCLFCYKRN protein nsp3-P1
RSLRVKLCSTI VGGMIRYYDVMANGGTGFCSKHQWNCIDCDSYK
PGNTFITVEAALDLSKELKRPIQPTD/ OC43- aa2542-2602/
NAAVFYAQLSLFRPILMVDKNLITTANTGTSVTETM nsp3-P2 aa2651-2705 FDVYVDTFSL
MFDVDKKSLNALIATA /ELTDESCNNLVPTYLKSDNIVAADLGVLIQNSAKH
VQGNVAKIAG VSCIWSVDAF Full OC43- aa2368-2473/
FMRFYIIIASFIFSLFRHVAYGCSKSGCLFCYKRN length nsp3-FL aa2542-2602/
RSLRVKLCSTIV aa2651-2705 GGMIRYYDVMANGGTGFCSKHQWNCIDCDSYK
GNTFITVEAA LDLSKELKRPIQPTD/NAAVFYAQLSLFRPILMV
DKNLITTANTGTSVTETMFDVYVDTFSLMFDVDKK SLNALIATA/
ELTDESCNNLVPTYLKSDNIVAADLGVLIQNSAKH VQGNVAKIAG VSCIWSVDAF Short
OC43- aa2875-2902 SADGVQCYTPHSQISYSNFIYASGCVLSS peptide nsp4-S1 OC43- aa2893-
2919 FYASGCVLSSACTMFMADGSPQPYCY/ nsp4-S2 OC43- aa2932-2957
SLVPHVRYNLANAKGFIRFPEVLREG nsp4-S3 OC43- aa2948-2973
IRFPEVLREGLVRIVRTRSMSYCRVG nsp4-S4 OC43- aa2964-2988
TRSMSYCRVGLCEEADEGICFNFNNG nsp4-S5 OC43- aa2979-3003
DEGICFNFNNGSWVLNNDYYRSLPGT nsp4-S6 OC43- aa2994-3018
NDYYRSLPGTFCGRDVFDFLIYQLFK nsp4-S7 OC43- aa3009-3033
VDFLIYQLFKGLAQPVDFLALTASS nsp4-S8 OC43- aa3024-3048
VDFLALTASSIAGAILAVIVVLVIFY nsp4-S9 OC43- aa3039-3064
LAVIVVLVIFYLIKLKRAFGDYTSVV/ nsp4-S10 OC43- aa3187-3216
NRYLSLYNKYRYYSYGKMDTAAYREAACSQL nsp4-S11 OC43- aa3207-3236
AYREAACSQLAKAMDTFTNNNGSDVLYQPP nsp4-S12 Extended OC43- aa2875-2919
SADGVQCYTPHSQISYSNFIYASGCVLSSACTMFTMADGSP peptide nsp4-E1 QPYCY OC43-
aa2932-2977 SLVPHVRYNLANAKGFIRFPEVLREGLVRIVRTRSM nsp4-E2 SYCRVGLCEE
OC43- aa2968-3011 SYCRVGLCEEADEGICFNFNNGSWVLNNDYYRSLPG nsp4-E3
TFCGRDVFDF OC43- aa3002-3046 GTFCGRDVFDFLIYQLFKGLAQPVDFLALTASSIAGA
nsp4-E4 ILAVIVVLV OC43- aa3038-3064/ ILAVIVVLVIFYLIKLKRAFGDYTSVV/ nsp4-E5
aa3187-3203 NRYLSLYNKYRYYSYGKM OC43- aa3195-3236
KYRYYSYGKMDTAAYREAACSQLAKAMDTFTNNN nsp4-E6 GSDVLYQPP Short OC43-
aa2875-2919/ SADGVQCYTPHSQISYSNFIYASGCVLSSACTMFTM protein nsp4-P1 aa2932-
3006 ADGSPQPYCY/ SLVPHVRYNLANAKGFIRFPEVLREGLVRIVRTRSM SYCRVGLCEE
ADEGICFNFNNGSWVLNNDYYRSLPGTFCG OC43- aa2996-3064/

YYRSLPTGTFCDRLYQFLFKGLAQPVDLFLALTA nsp4-P2 aa3187-3236 SSIAGAILA
 VIVVLVFYYLIKLKRAFGDYTSVV/ NRYLSLYNKYRYYSKGMDTAAYREAACSQLAKA
 MDTFTNNN GSDVLYQPP Full OC43- aa2875-2919/
 SADGVQCYPHSPHSQISYSNFYASGCVLSSACTMFTM length nsp4-FL aa2932-3064/
 ADGSPQPYCY/ aa3187-3236 SLVPHVRYNLANAKGFIRFPEVLREGLVRIVRTRSM
 SYCRVGLCEEA DEGIC FNFNGSWVLNNDYRSLPGTFCGRDVFDFLIYQLFK
 GLAQPVDLFLAL TASSIA GAILAVIVVLVFYYLIKLKRAFGDYTSVV/
 NRYLSLYNKYRYYSKGMDTAAYREAACSQLAKA MDTFTNNN GSDVLYQPP .sup.1aa
 positions based on OC43 reference ORF1ab protein sequence, YP_009555238.1 .sup.2For
 conjugation to polyionic VLPS, the peptide/protein antigens have a N-terminal TAG and AAYY
 proteolytic processing sequence. / indicates break in native amino acid sequence
 TABLE-US-00015 TABLE 15 OC43 antigens of nonstructural proteins nsp6 and
 nsp7 Immunogen Seq ID aa positions.sup.1 Amino acid sequence.sup.2 Short OC43-nsp6-
 S1 aa3602-3622 SLAMLLVKHKHLYLTMYITPV peptide OC43-nsp6-S2 aa3613-3633
 LYLTMYITPVLFLLYNNYL LV OC43-nsp6-S3 aa3753-3780
 IKIVLLCYLFIGYIISCYWGLFSLMNSL OC43-nsp6-S4 aa3770-3798
 WGLFSLMNSLFRMPLGVYNYKISVQELRY OC43-nsp6-S5 aa3789-3816
 YKISVQELRYMNANGLRPPKNSFEALML OC43-nsp6-S6 aa3807-3836
 PKNSFEALMLNFKLLGIGGVPIIEVSQFQ Extended OC43-nsp6-E1 aa3602-3633
 SLAMLLVKHKHLYLTMYITPVLFLLYNNY peptide LV OC43-nsp6-E2 aa3753-3799
 IKIVLLCYLFIGYIISCYWGLFSLMNSLFRMPL GVYNYKISVQELRY OC43-nsp6-E3 aa3790-
 3836 YKISVQELRYMNANGLRPPKNSFEALMLNF KLLGIGGVPIIEVSQFQ Short OC43-
 nsp6-P1 aa3602-3633/ SLAMLLVKHKHLYLTMYITPVLFLLYNNY protein aa3753-3836 LV/
 IKIVLLCYLFIGYIISCYWGLFSLMNSLFRMPL
 GVYNYKISVQELRYMNANGLRPPKNSFEAL MLNFKLLGIGGVPIIEVSQFQ Short OC43-
 nsp7-S1 aa3837-3862 SKLTDVKCANVVLLNCLQHLHVASNS peptide OC43-nsp7-S2 aa3853-
 3878 LQHLHVASNSKLWHYCSTLHNEILAT OC43-nsp7-S3 aa3869-3894
 STLHNEILATSDLSVAFEKLAQLLIV OC43-nsp7-S4 aa3885-3910
 FEKLAQLLIVLFANPAAVDSKCLTSI OC43-nsp7-S5 aa3901-3925
 AVDSKCLTSIEEVCDDYAKDNTVLQ Extended OC43-nsp-7- aa3837-3875
 SKLTDVKCANVVLLNCLQHLHVASNSKLW peptide E1 HYCSTLHNEI OC43-nsp-7- aa3866-
 3903 HYCSTLHNEILATSDLSVAFEKLAQLLIVLF E2 ANPAAVD Short OC43-nsp7-P1
 aa3837-3925 SKLTDVKCANVVLLNCLQHLHVASNSKLW protein
 HYCSTLHNEILATSDLSVAFEKLAQLLIVLF ANPAAVDSKCLTSIEEVCDDYAKDNTVLQ
 .sup.1aa positions based on OC43 reference ORF1ab protein sequence, YP_009555238.1 .sup.2For
 conjugation to polyionic VLPS, the peptide/protein antigens have a N-terminal TAG and AAYY
 proteolytic processing sequence. / indicates break in native amino acid sequence
 TABLE-US-00016 TABLE 16 OC43 antigens homologous to the nsp12-1
 nonstructural protein fragment of SARS-CoV-2 Immunogen Seq ID aa-position.sup.1
 Amino acid sequence.sup.2 Short OC43 aa4525-4555
 YFTKKDWYDFVENPDIINVYKKLGPIFNRL peptide nsp12-1-S1 OC43 aa4546-4575
 KLGPIFNRLVSATEFADKLVEVGLVGVLTL nsp12-1-S2 OC43 aa4566-4594
 VEVGLVGVLTLNQLNGKWYDFGDYVIA nsp12-1-S3 OC43 aa4585-4615
 WYDFGDYVIAAPGCGVAIADSYYSYIMPMLT nsp12-1-S4 SARS- aa4634-4663
 DLVQYDFTDYKLELFNKYFKHWSMPYHPNT CoV-2 nsp12-1-S5 OC43 aa4654-4683
 HWSMPYHPNTVDCQDDRCIIHCANFNILFS nsp12-1-S6 OC43 aa4674-4703
 HCANFNILFSMVLPNTCFGPLVRQIFVDG nsp12-1-S7 OC43 aa4693-4721
 PLVRQIFVDGVVPFVVSIGYHYKELGIVMN nsp12-1-S8 OC43 aa4712-4740
 HYKELGIVMNMDVDTHRYRLSLKDLLLLYA nsp12-1-S9 Extended OC43 aa4525-4574
 YFTKKDWYDFVENPDIINVYKKLGPIFNRLVSATEFAD peptide nsp12-1-E1

KLVEVGLVGVLTLDNQDLNGKWYDFGDYVIAAPGCGVAIA nsp12-1-E2 DSYYSYIMPMLT
OC43 aa4634-4675 DLVQYDFTDYKLELFNKYFKHWSMPYHPNTVDCQDDRCI nsp12-1-E3
IHC OC43 aa4666-4707 CQDDRCIIHCANFNILFSMVLNPNTCFGPLVRQIFVDGVP nsp12-1-E4
FVV OC43 aa4698-4740 IFVDGVPFVVSIGYHYKELGIVMNMDVDTHRYRLSLKDL nsp12-1-
E5 LLYA Short OC43 aa4525-4615 YFTKKDWYDFVENPDIINVYKKLGPIFNRALVSATEFAD
protein nsp12-1- KLVEVGLVGVLTLDNQDLNGKWYDFGDYVIAAPGCGVAI SP1
ADSYYSYIMPMLT OC43 aa4634-4740
DLVQYDFTDYKLELFNKYFKHWSMPYHPNTVDCQDDRCI nsp12-1-
IHCANFNILFSMVLNPNTCFGPLVRQIFVDGVPFVVSIGY SP2
HYKELGIVMNMDVDTHRYRLSLKDLLLYA Full OC43 aa4525-
YFTKKDWYDFVENPDIINVYKKLGPIFNRALVSATEFAD length nsp12-1-FL 4615/
KLVEVGLVGVLTLDNQDLNGKWYDFGDYVIAAPGCGVAI aa4634-
ADSYYSYIMPMLT/DLVQYDFTDYKLELFNKYFKHWSMP 4740
YHPNTVDCQDDRCIIHCANFNILFSMVLNPNTCFGPLVRQ
IFVDGVPFVVSIGYHYKELGIVMNMDVDTHRYRLSLKDL LLYA .sup.laa positions based on
OC43 reference ORF1ab protein (YP_009555238.1). .sup.2For conjugation to polyionic VLPS, the
peptide/protein antigens have a N-terminal TAG and AAYY proteolytic processing sequence. /
indicates break in native amino acid sequence

TABLE-US-00017 TABLE 17 OC43 antigens homologous to of the nsp12-2
nonstructural protein fragment of SARS-CoV-2 Immunogen Seq ID aa-position Amino
acid sequence Short OC43 aa4894-4921 AYTKRNVLP TLTQMNLKYAISAKNRART peptide
nsp12-2- S1 OC43 aa4912-4939 AISAKNRARTVAGVSILSTMTGRMFHQK nsp12-2- S2 OC43
aa4930-4957 TMTGRMFHQKCLKSIAATRGVPVIGTT nsp12-2- S3 OC43 aa4948-4975
RGVPVIGTTKFYGGWDDMLRRLIKDVD nsp12-2- S4 OC43 aa4966-4993
MLRRLIKDVDPVLMGWDYPKCDRAMPN nsp12-2- S5 OC43 aa4984-5012
YPKCDRAMPNLLRIVSSLVLARKHETCC nsp12-2- S S6 OC43 aa5003-5030
LARKHETCCSQSDRFYRLANECAQVLSE nsp12-2- S7 OC43 aa5021-5048
ANECAQVLSEIVMCGGCYVVKPGGTSSG nsp12-2- S8 OC43 aa5039-5067
YVKPGGTSSGDATTAFAANSVFNICQAVS nsp12-2- A S9 OC43 aa5058-5085
VFNICQAVSANVCALMSCNGNKIEDLSI nsp12-2- S10 OC43 aa5076-5103
NGNKIEDLSIRALQKRLYSHVYRSDKVD nsp12-2- S11 OC43 aa5094-5121
SHVYRSDKVDSTFVTEYYEFLNKHFSMM nsp12-2- S12 OC43 aa5112-5140
EFLNKHFSMMILSDDGVVCYNSDYASKG nsp12-2- Y S13 OC43 aa5131-5159
YNSDYASKGYIANISAFQQVLYYQNNVF nsp12-2- M S14 OC43 aa5150-5177
VLYYQNNVFMSESKCWVEHDINNGPHEF nsp12-2- S15 OC43 aa5168-5196
HDINNGPHEFCSQHTMLVKMDGDDVYLP nsp12-2- Y S16 OC43 aa5187-5215
MDGDDVYLPYPNPSRILGAGCFVDDLLK nsp12-2- T S17 OC43 aa5206-5234
GCFVDDLLKTDSVLLIERFVSLAIDAYP nsp12-2- L S18 Extended OC43 aa4894-4940
AYTKRNVLP TLTQMNLKYAISAKNRART peptide nsp12-2- VAGVSILSTMTGRMFHQK E1
OC43 aa4931-4977 MTGRMFHQKCLKSIAATRGVPVIGTTK nsp12-2-
FYGGWDDMLRRLIKDVDPNP E2 OC43 aa4968-5013
RRLIKDVDPVLMGWDYPKCDRAMPN nsp12-2- LLRIVSSLVLARKHETCCSQ E3 OC43
aa5004-5049 ARKHETCCSQSDRFYRLANECAQVLSEI nsp12-2-
VMCGGCYVVKPGGTSSGD E4 OC43 aa5040-5085
VKPGGTSSGDATTAFAANSVFNICQAVSA nsp12-2- NVCALMSCNGNKIEDLSI E5 OC43
aa5076-5121 NGNKIEDLSIRALQKRLYSHVYRSDKVDS nsp12-2- TFVTEYYEFLNKHFSMM
E6 OC43 aa5112-5158 EFLNKHFSMMILSDDGVVCYNSDYASKG nsp12-2-
YIANISAFQQVLYYQNNVF E7 OC43 aa5149-5195 QVLYYQNNVFMSESKCWVEHDINNGP
nsp12-2- HEFCSQHTMLVKMDGDDVYLP E8 OC43 aa5188-5234

KMDGDDVLYPYNPSRILGACFVDDLL nsp12-2- KTDSVLLIERFVSLAIDAYPL E9 Short
 OC43 aa4894-5018 AYTKRNVLPPTLTQMNLKYAISAKNRART protein nsp12-2-
 VAGVSILSTMTGRMFHQKCLKSIAATRG SP1 VPVIGTTKFYGGWDDMLRRLIKDVDN
 PVLMGWDYPKCDRAMPNLLRIVSSLVL ARKHETCCSQSDRFY OC43 aa5008-5132
 ETCCSQSDRFYRLANECAQVLSEIVMCG nsp12-2-
 GCYYVKPGGTSSGDATTAFAANSVFNICQ SP2 AVSANVCALMSCNGNKIEDLSIRALQKR
 LYSHVYRSDKVDSTFVTEYYEFLNKHFS MMILSDDGVVCYN OC43 aa5122-5234
 ILSDDGVVCYNNDYASKGYIANISAFQQV nsp12-2-
 LYYQNNVFMSESKCWVEHDINNGPHEF SP3 CSQHTMLVKMDGDDVYLPYPNPSRILG
 AGCFVDDLLKTDSVLLIERFVSLAIDAYP L Full OC43 aa4894-5234
 AYTKRNVLPPTLTQMNLKYAISAKNRART length nsp12-2-
 VAGVSILSTMTGRMFHQKCLKSIAATRG antigen FL
 VPVIGTTKFYGGWDDMLRRLIKDVDN PVLMGWDYPKCDRAMPNLLRIVSSLVL
 ARKHETCCSQSDRFYRLANECAQVLSEI VMCGGCYYVKPGGTSSGDATTAFAANSV
 FNICQAVSANVCALMSCNGNKIEDLSIR ALQKRLYSHVYRSDKVDSTFVTEYYEFL
 NKHFSMMILSDDGVVCYNNDYASKGYIA NISAFQQVLYYYQNNVFMSESKCWVEHDI
 NNGPHEFCSQHTMLVKMDGDDVYLPY PNPSRILGAGCFVDDLLKTDSVLLIERFVS
 LAIDAYPL .sup.1aa positions based on OC43 reference ORF1ab protein (YP_009555238.1).
 .sup.2For conjugation to polyionic VLPS, the peptide/protein antigens have a N-terminal TAG and
 AAYY proteolytic processing sequence.

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Claims

1. A composition comprising (a) a chimeric papillomavirus virus-like particle (VLP) comprising an L1 protein, having an amino acid insert inserted into the HI loop of the L1 protein and (b) a coronavirus antigen.
2. A composition according to claim 1 wherein said amino acid insert comprises a contiguous sequence of negatively charged amino acids and a terminal cysteine residue.
3. (canceled)
4. (canceled)
5. A composition according to claim 2, wherein said amino acid insert is selected from the inserts identified in Table 1.
6. A composition according to claim 1, wherein said amino acid insert is inserted into the HI loop of the L1 protein at a location between positions 344 and 357.
7. (canceled)
8. A composition according to claim 6, wherein said amino acid insert replaces the native amino acid sequences identified in Table 1.
9. A composition according to claim 1, wherein said coronavirus antigen is selected from the group consisting of an OC43 antigen, an HKU1 antigen, a 229E antigen, an NL63 antigen, a SARS-CoV-1 antigen, a MERS antigen, a SARS-CoV-2 antigen, and fusions thereof.
10. (canceled)
11. A composition according to claim 9, wherein said coronavirus antigen comprises a viral structural protein selected from the group consisting of the membrane protein (M), the nucleocapsid protein (N), and the S2 region of the spike(S) envelope protein from OC43, HKU1, 229E, NL63, SARS-CoV-1, MERS, and/or SARS-CoV-2 coronaviruses or a viral non-structural

protein selected from the group consisting of nsp3, nsp4, nsp6, nsp7, and nsp12 proteins from OC43, HKU1, 229E, NL63, SARS-CoV-1, MERS, and/or SARS-CoV-2 coronaviruses.

12. (canceled)

13. (canceled)

14. (canceled)

15. A composition according to claim 9, wherein said coronavirus antigen is selected from one or more of the amino acid sequences in Tables 3-10.

16. (canceled)

17. (canceled)

18. (canceled)

19. (canceled)

20. (canceled)

21. (canceled)

22. (canceled)

23. A composition according to claim 9, wherein said coronavirus antigen comprises one or more viral proteins of the OC43 coronavirus and/or HKY1 the coronavirus, wherein said one or more viral proteins comprise 70% or greater identity with an amino acid sequence of SARS-CoV-1 coronavirus, MERS coronavirus, and/or SARS-CoV-2 coronavirus.

24. (canceled)

25. A composition according to claim 9, wherein said OC43 and/or HKU1 coronavirus structural protein antigens is selected from one or more of the amino acid sequences in Tables 11-13 and/or said OC43 coronavirus nonstructural antigen is selected from one or more of the amino acid sequences in Tables 14-17.

26. (canceled)

27. (canceled)

28. (canceled)

29. A composition according to claim 9, wherein said coronavirus antigen comprises a first antigen selected from the structural proteins and peptides of OC43 and/or HKU1 coronavirus, a second antigen selected from the non-structural proteins and peptides of OC43, a third antigen selected from the structural proteins and peptides of SARS-CoV-2 coronavirus, and a fourth antigen selected from the nonstructural proteins and peptides of SARS-CoV-2.

30. A composition according to claim 1, further comprising a TAG sequence linked at a first end to said amino acid insert and linked at a second end to said coronavirus antigen.

31. (canceled)

32. (canceled)

33. (canceled)

34. A composition according to claim 30, wherein said TAG sequence comprises an amino acid sequence identified in Table 2.

35. A composition according to claim 1, wherein said composition is effective to stimulate a cytotoxic T cell response in a mammal.

36. A method for stimulating a cytotoxic T cell response to a coronavirus in a mammal or for stimulating both therapeutic and protective immunity to a coronavirus in said mammal comprising administering to said mammal a composition according to any one of claims 1-14.

37. (canceled)

38. (canceled)

39. (canceled)

40. (canceled)

41. (canceled)

42. (canceled)

43. (canceled)

44. (canceled)
45. (canceled)
